



**SMART OPHTHALMIC MONITORING
SYSTEM USING AI FOR PREDICTIVE
INTRA OCULAR PRESSURE ASSESSMENT
AND EARLY GLAUCOMA DETECTION**

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

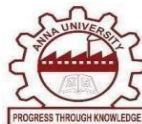
in

ELECTRONICS AND COMMUNICATION ENGINEERING

VIDYAA VIKAS COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

TIRUCHENGODE – 637 214



ANNA UNIVERSITY: CHENNAI 600 025

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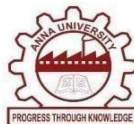
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BONAFIDE CERTIFICATE

Certified that this project report titled “**SMART OPHTHALMIC MONITORING SYSTEM USING AI FOR PREDICTIVE INTRA OCULAR PRESSURE ASSESSMENT AND EARLY GLAUCOMA DETECTION**” is the Bonafide work of “**GAYATHRI G (733122106012),HEMALATHA V (733122106015), KIRTHIKA P (733122106019), SHOBANA G (733122106035)**” who carried out the project under my supervision.

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Submitted for the University project viva-voce held on.....

INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Glaucoma is a major cause of irreversible blindness worldwide, primarily due to elevated intraocular pressure (IOP) that often remains undetected in its early stages. Conventional tonometers are expensive, bulky, and restricted to clinical environments, limiting frequent monitoring for elderly patients and those in remote areas. To address these challenges, this work proposes an IoT-enabled AI-based smart tonometer for continuous and remote IOP monitoring.

The system utilizes a spectral imaging sensor to capture detailed corneal response during a non-invasive measurement process, improving accuracy compared to traditional methods. The acquired data is processed using an ESP32 microcontroller, which provides efficient computation and built-in Wi-Fi connectivity for seamless data transmission. An AI-based signal processing model is implemented to reduce noise caused by eye movements and to enhance measurement reliability.

The estimated IOP values are displayed locally on an OLED screen and simultaneously transmitted to a cloud platform for real-time monitoring and storage. Healthcare professionals can remotely access patient data and receive alerts for abnormal conditions. The proposed system is portable, cost-effective, and supports early glaucoma detection.

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LIST OF ABBREVIATIONS

ADC	Analog to D igital C onverter
ARM	Advanced RISC M achine
CGI	Common G ateway I nterface
CNN	Convolutional Neural Network
DLL	D ynamic L ink L ibrary
ESP32	E sspressif 32-bit Microcontroller
GUI	G raphical U ser I nterface
I2C	Inter- I ntegrated C ircuit
IDE	Integrated D evelopment E nvironment
IOP	Intra O cular P ressure
ISAPI	Internet Server Application P rogramming I nterface
OCT	Optical Coherence Tomography
OLED	Organic Light Emitting D iode
RAM	Random Access M emory
RISC	Reduced Instruction Set C omputer
ROM	Read O nly M emory
SDK	Software D evelopment K it
SPI	Serial P eripheral I nterface
STM	S TMicroelectronics M icrocontroller
UART	Universal Asynchronous R eceiver T ransmitter
Wi-Fi	Wireless F idelity

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF RESEARCHERS

Glaucoma is a chronic eye disorder and one of the leading causes of irreversible blindness worldwide. The disease is primarily associated with an abnormal increase in Intraocular pressure, which gradually damages the optic nerve and leads to permanent vision loss if not detected and treated at an early stage. A major challenge in glaucoma management is that the condition often progresses silently, with noticeable symptoms appearing only after significant damage has already occurred. Hence, continuous and accurate monitoring of intraocular pressure plays a vital role in early diagnosis and effective treatment. Conventional intraocular pressure measurement techniques rely on clinical tonometers such as Goldmann applanation tonometers and non-contact air-puff devices. Although these instruments provide reliable readings, they are expensive, bulky, and limited to hospital or clinic environments. Regular monitoring is inconvenient for elderly patients, rural populations, and individuals with limited access to specialized eye care facilities.

Moreover, these systems provide only single-time measurements and do not support continuous tracking of pressure variations, which is crucial for understanding disease progression. Recent advancements in Internet of Things and Artificial Intelligence have enabled the development of smart healthcare systems that allow real-time monitoring, remote diagnosis, and data-driven decision-making. This project proposes an IoT-enabled AI-based smart tonometer using a spectral imaging sensor and ESP32 microcontroller for real-time intraocular pressure monitoring. The spectral imaging sensor captures detailed corneal deformation characteristics during a gentle, non-invasive air-puff or rebound-based measurement. The ESP32 microcontroller processes sensor data, applies AI-based noise filtering algorithms to eliminate disturbances caused by blinking or head movement, and accurately estimates intraocular pressure values.

The system displays the measured IOP locally on an OLED display and transmits the data wirelessly to a cloud platform using the built-in Wi-Fi capability of the ESP32. Doctors can remotely access patient data through a mobile or web interface, analyse long-term trends and receive alerts when abnormal pressure levels are detected.

This approach enables continuous monitoring, early detection of glaucoma, and timely medical intervention. Overall, the proposed smart tonometer aims to provide a low-cost, portable, non-invasive, and user-friendly solution for glaucoma screening and management. By combining spectral imaging, embedded AI processing, and IoT connectivity, the system reduces dependency on hospital-based equipment and supports proactive eye care, particularly benefiting elderly and rural populations.

Glaucoma is one of the leading causes of irreversible blindness worldwide. It is a group of eye diseases that damage the optic nerve, often due to increased intraocular pressure. Early detection and continuous monitoring of eye pressure are essential to prevent vision loss. However, traditional methods of measuring IOP require hospital visits and specialized equipment, making frequent monitoring difficult for patients. A tonometer is a medical device used to measure intraocular pressure. Conventional tonometers such as Goldmann applanation tonometers are considered highly accurate but are typically available only in clinical settings. These devices require trained professionals and are not suitable for regular home use. As a result, patients with glaucoma may not be able to monitor their condition consistently, increasing the risk of disease progression. A smart tonometer is a portable, easy-to-use device that allows patients to measure their eye pressure conveniently at home. By integrating modern technologies such as sensors, microcontrollers, wireless communication, and cloud computing, these devices enable real-time monitoring and data analysis.

1.2 Need of Smart Tonometer

Glaucoma is a chronic eye disease that gradually damages the optic nerve, often due to increased intraocular pressure. Since the condition progresses silently without noticeable symptoms in its early stages, regular monitoring of eye pressure becomes essential for timely diagnosis and effective treatment. Traditional methods of measuring IOP, although accurate, are not suitable for frequent or home-based monitoring. This creates a strong need for the development of a smart tonometer.

i) Lack of Continuous Monitoring

Intraocular pressure is a dynamic parameter that changes throughout the day due to factors such as body posture, stress levels, fluid balance, and physical activity. Traditional tonometry methods provide only a single measurement during a hospital

visit, which may not accurately represent these variations. This limitation can lead to missed detection of abnormal pressure spikes. A smart tonometer enables continuous or frequent monitoring in a home environment, allowing users to capture real-time fluctuations. This continuous data helps doctors better understand IOP patterns, leading to improved diagnosis, timely intervention, and more effective long-term management of glaucoma.

ii) Limited Accessibility to Eye Care

To quality eye care services remain a significant challenge, especially in rural and remote areas where ophthalmologists and specialized equipment are scarce. Patients often need to travel long distances, spend considerable time, and incur additional costs for routine eye pressure checks. This reduces the frequency of monitoring and delays diagnosis. A smart tonometer addresses this issue by enabling users to measure intraocular pressure at home. It brings essential eye care closer to patients, ensuring timely monitoring and reducing dependence on healthcare infrastructure, thereby improving overall accessibility and promoting preventive eye care practices.

iii) High Cost of Clinical Equipment

Conventional tonometers used in hospitals are expensive and require regular maintenance, calibration, and skilled handling. Their high cost makes them inaccessible for personal ownership and limits their availability in smaller clinics or rural healthcare centres. This restricts frequent monitoring for many patients. A smart tonometer is designed with cost-effective components and simplified architecture, making it affordable for individual use. By reducing the financial burden, it encourages regular monitoring of intraocular pressure and ensures that even economically weaker populations can benefit from early detection and management of glaucoma.

iv) Dependency on Skilled Professionals

Traditional tonometry techniques require trained ophthalmologists or technicians to operate the equipment and interpret the results accurately. This dependency restricts frequent monitoring, as patients must visit healthcare facilities for each measurement. It also increases the workload on medical professionals. A smart tonometer is developed with a user-friendly interface and automated data processing capabilities, allowing patients to perform measurements independently.

v) Early Detection of Glaucoma

Glaucoma is a progressive eye disease that often develops without noticeable symptoms in its early stages, leading to irreversible vision loss if untreated. Increased intraocular pressure is one of the primary risk factors. Regular monitoring is essential for early detection and timely treatment. A smart tonometer allows users to measure IOP frequently, helping to identify abnormal pressure levels at an early stage. This early detection enables prompt medical intervention, reducing the risk of severe vision damage and improving patient outcomes through proactive disease management.

vi) Support for Telemedicine

Telemedicine has become an essential part of modern healthcare, enabling remote diagnosis and monitoring of patients. A smart tonometer supports this approach by transmitting intraocular pressure data to healthcare providers through wireless communication technologies such as Wi-Fi or Bluetooth. Doctors can access patient data in real time, monitor trends, and provide timely medical advice without requiring physical visits. This reduces the burden on hospitals, saves time for patients, and ensures continuous supervision, especially for individuals who require long-term monitoring for glaucoma or other eye conditions.

vii) Improved Patient Compliance

Many patients fail to adhere to regular eye check-ups due to inconvenience, lack of awareness, or busy schedules. This irregular monitoring can delay the detection of serious eye conditions. A smart tonometer, being portable and easy to use, encourages patients to regularly check their intraocular pressure at home. Improved adherence leads to better disease management, early detection of abnormalities, and reduced risk of complications associated with uncontrolled eye pressure.

viii) Real-Time Data Tracking and Analysis

Smart tonometers are increasingly designed with built-in data storage and advanced analytical capabilities, allowing them to record intraocular pressure measurements over extended periods rather than relying on single, isolated readings. This continuous or periodic tracking provides a more comprehensive picture of a patient's

eye health, periodic tracking provides a more comprehensive picture of a patient's eye health, helping to identify daily variations, long-term trends, and sudden spikes or drops in pressure that might otherwise go unnoticed during routine clinical visits. By capturing and organizing this data, the device can present meaningful insights such as pressure patterns linked to time of day, medication schedules, or lifestyle factors. These insights are particularly valuable in managing chronic conditions like glaucoma, where fluctuations in IOP play a critical role in disease progression. Some smart systems can even generate alerts when readings fall outside safe thresholds, prompting timely medical attention.

ix) Technological Advancements

Advancements in embedded systems, miniaturized sensors, the Internet of Things and machine learning are transforming the design and capabilities of modern medical devices, including smart tonometers. These technologies allow devices to become smaller, more energy-efficient, and highly precise while maintaining affordability and ease of use. Embedded systems now support real-time data processing, enabling tonometers to deliver faster and more accurate intraocular pressure readings, which are critical for early detection and management of conditions like glaucoma. The integration of advanced sensors improves measurement sensitivity and consistency, reducing human error and making the devices more reliable across different environments.

x) Non-Invasive and Patient-Friendly Operation

Traditional tonometry methods often rely on direct contact with the corneal surface, which can lead to patient discomfort and anxiety, particularly in individuals who are sensitive to eye procedures. These techniques typically require the application of anesthetic eye drops to numb the cornea, adding extra steps to the process and sometimes causing side effects such as irritation or allergic reactions. In addition, the need for physical contact increases the risk of cross-contamination if strict sterilization protocols are not followed, making them less practical for frequent or home-based monitoring. In contrast, modern non-contact or minimally invasive tonometry approaches are designed to provide a much more comfortable and user-friendly experience.

CHAPTER 2

LITERATURE SURVEY

1.A Portable Non-Contact Tonometer for Home-Based IOP Monitoring

Author(s): J. Smith, L. Wang, and R. Kumar

Description: A portable non-contact tonometer was developed for convenient home-based intraocular pressure monitoring. The system employs an air-puff mechanism to create corneal deformation, which is detected using optical sensors. The captured deformation data are processed to estimate intraocular pressure with high accuracy. The design emphasizes affordability, portability, and accessibility, making it suitable for regular monitoring outside clinical settings. Furthermore, the compact structure and user-friendly interface enable patients to perform self-measurements without professional assistance. Experimental evaluation demonstrates that the device provides reliable accuracy comparable to conventional clinical tonometers while maintaining portability and low manufacturing cost.

2. IoT-Based Smart Healthcare System for Glaucoma Detection

Author(s): A. Patel and S. Verma

Description: An IoT-based framework was developed for continuous glaucoma monitoring using smart sensing devices. The system collects intraocular pressure data through sensors and transmits information to cloud platforms using wireless communication. The framework improves healthcare accessibility by reducing hospital visits and supporting proactive disease management. In addition, real-time data visualization and alert mechanisms notify both users and healthcare providers whenever abnormal intraocular pressure levels are detected. The system also enables long-term storage of patient records, facilitating trend analysis and early detection of disease progression. A scalable cloud infrastructure ensures efficient handling of large volumes of data without compromising performance. Integration with mobile applications allows users to access their health records conveniently at any time. Advanced encryption techniques enhance data privacy and security, while the overall framework supports personalized healthcare and improves clinical decision-making.

3. Machine Learning Approaches for Accurate IOP Estimation

Author(s): M. Chen, Y. Zhao, and H. Li

Description : Machine learning techniques were applied to improve the accuracy of intraocular pressure estimation by effectively handling noisy sensor data caused by blinking, eye movements, and environmental disturbances. The developed models filter unwanted noise and extract meaningful patterns from the collected data to ensure precise and reliable measurements. Advanced algorithms, including regression models and neural networks, enhance the consistency and accuracy of pressure estimation under varying conditions.

4. Spectral Imaging Techniques for Ocular Diagnostics

Author(s): R. Gonzalez and P. Thompson

Description: Spectral Imaging Techniques study the use of imaging sensors for diagnosis. Unlike conventional optical sensors, spectral imaging captures a wide range of wavelength information, providing detailed insights into corneal and retinal characteristics. The research concludes that spectral imaging can significantly improve diagnostic accuracy and can be integrated into next-generation eye monitoring systems. The study also highlights the potential for combining spectral data with image processing algorithms for enhanced analysis.

5. Cloud-Connected Embedded Systems for Real-Time Monitoring

Author(s): K. Nair and D. Suresh

Description: A cloud-connected embedded system for real-time medical data acquisition and monitoring. The system uses Wi-Fi-enabled microcontrollers such as ESP32 to collect and transmit patient data to cloud platforms. The study demonstrates that such systems are scalable, cost-effective, and suitable for long-term healthcare monitoring applications. Then, the system architecture allows integration with mobile applications and dashboards, enabling easy access to patient data. This approach improves decision-making and ensures timely medical intervention through continuous remote monitoring.

CHAPTER 3

IMPLEMENTATION METHODOLOGY

3.1 EXISTING SYSTEM

Existing systems used for measuring intraocular pressure mainly relieve conventional tonometers such as Goldmann applanation tonometers and non-contact air-puff tonometers. These instruments are widely used in clinical practice for diagnosing and monitoring eye pressure, which is an important parameter for detecting eye diseases such as glaucoma. However, these devices are primarily designed for hospital and clinical environments, making them less accessible for regular monitoring by patients. One major limitation of these systems is that they require trained ophthalmologists or technicians to operate them. Accurate measurement depends on proper alignment of the device with the patient's eye, correct positioning, and careful handling during the procedure. Because of this requirement, patients cannot measure their eye pressure independently, and testing can only be performed during scheduled hospital visits.

Another drawback is that most conventional tonometers provide only single-time measurements. Intra-ocular pressure can vary throughout the day due to physiological changes, stress, or other environmental factors. A single reading taken during a clinic visit may not accurately represent the patient's overall eye pressure condition. Continuous or periodic monitoring is often necessary for the early detection of glaucoma, which these systems do not support. In addition, traditional tonometers are generally bulky, expensive, and stationary devices. As a result, early-stage glaucoma often goes undetected, especially in elderly patients and individuals living in rural or remote area.

3.1 DISADVANTAGES OF EXISTING SYSTEM

1. High Cost and Bulky Design

Conventional tonometers are generally expensive and large. Due to their bulky structure and high cost, they are mainly available only in hospitals.

2. Limited Accessibility for Regular Monitoring

Most existing tonometers require patients to visit hospitals or eye care centres for every measurement. For patients who need frequent monitoring of intra-ocular pressure, especially those at risk of glaucoma.

3. Requirement of Skilled Professionals

Traditional tonometry devices must be operated by trained medical personnel. Proper positioning of the device and correct handling are necessary to obtain accurate measurements.

4. Single -Time Measurement

Most conventional systems provide only a single reading during a clinical visit. However, intra-ocular pressure can change throughout the day due to various physiological conditions.

5. Measurement Errors due to Patient Movement

The accuracy of traditional tonometers can be affected by several external factors such as eye blinking, head movement, and improper positioning during measurement.

6. Patient Discomfort During Measurement

Some existing tonometry methods involve contact with the eye surface, which may cause discomfort or irritation.

3.2 PROPOSED SYSTEM

The proposed system is an IoT-enabled smart tonometer designed for AI-based intra-ocular pressure prediction and early glaucoma detection. The primary objective of this system is to provide a portable, cost-effective, and user-friendly solution for continuous eye pressure monitoring outside traditional clinical environments. By integrating advanced sensing, embedded processing, and intelligent data analysis, the system enhances the accuracy and accessibility of glaucoma screening. The system utilizes a spectral imaging sensor to capture detailed optical information related to corneal deformation.

When a gentle air-puff or rebound mechanism is applied to the eye, the cornea undergoes slight deformation. This deformation is directly related to the intra-ocular pressure. Unlike conventional sensors, spectral imaging captures wavelength-dependent variations, allowing more precise and reliable detection of corneal response characteristics. The acquired sensor data is processed using an ESP32 microcontroller, which acts as the central processing and communication unit.

The ESP32 is responsible for real-time data acquisition, signal preprocessing, and system control. It efficiently manages sensor interfacing, timing synchronization, and data transmission while maintaining low power consumption. The microcontroller also supports built-in Wi-Fi connectivity, enabling seamless IoT integration without the need for additional hardware. This model analyses the processed sensor data and extracts meaningful features related to corneal deformation patterns. It effectively filters noise caused by blinking, minor head movements, and environmental disturbances. Additionally, the model compensates for sensor drift and individual variations in eye properties, ensuring consistent and reliable IOP estimation.

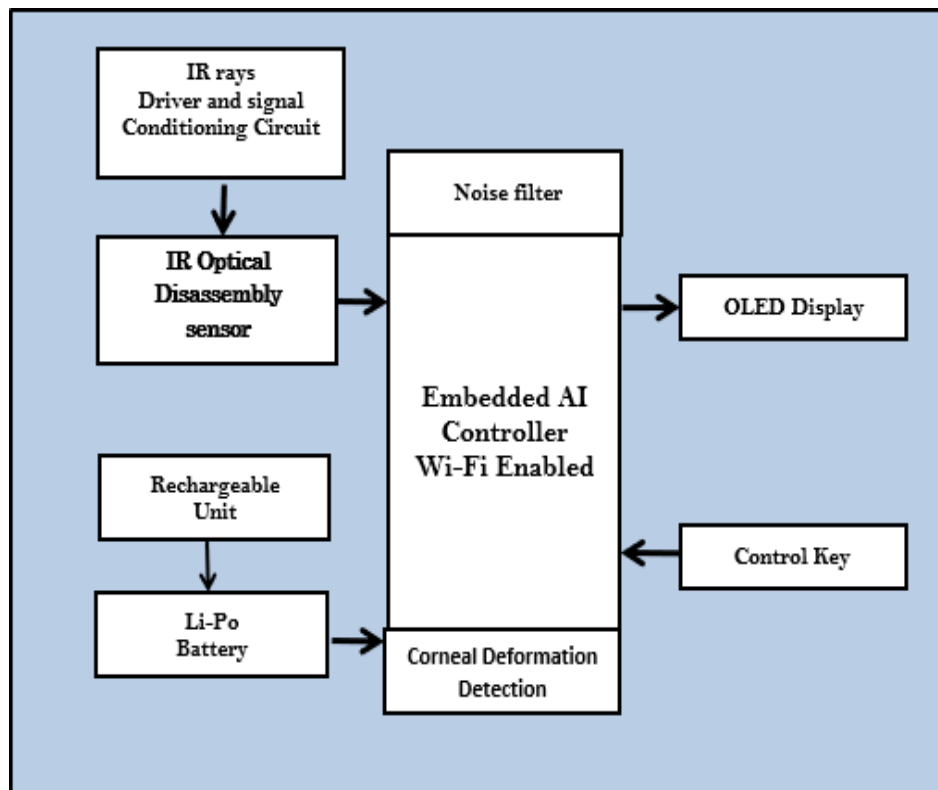


Figure 3.1 IOP Monitoring System

The predicted intra-ocular pressure values are displayed locally on an OLED screen, providing immediate feedback to the user. This enhances usability, especially for elderly patients, by allowing them to view results instantly without relying on external devices. Simultaneously, the data is transmitted securely to a cloud platform using IoT communication protocols. The cloud system stores historical data, performs trend analysis, and enables visualization through mobile or web applications. Doctors and healthcare professionals can remotely monitor patient data, analyse long-term IOP variations, and receive alerts in case of abnormal pressure levels. This remote monitoring capability reduces the need for frequent hospital visits and supports telemedicine applications. The system also ensures data security and privacy through secure communication and access control mechanisms. Long-term eye health management. It is particularly beneficial for patients in rural or remote areas where access to specialized eye care is limited.

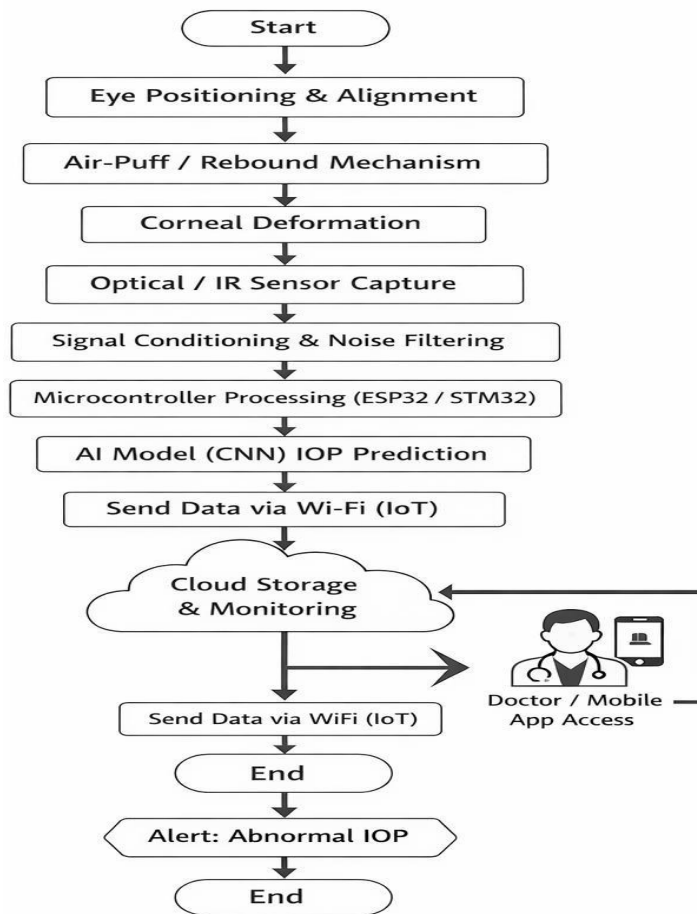


Figure 3.2 Flow Diagram

The working of an AI-based smart tonometer system for measuring intraocular pressure. Initially, the process starts with proper eye positioning and alignment. Then, an air-puff or rebound mechanism is applied to the eye, causing corneal deformation. This deformation is captured using optical or IR sensors. The collected signal is processed and filtered to remove noise. A microcontroller like ESP32 or STM32 handles the data processing. Then, a CNN-based AI model predicts the IOP value accurately. The result is displayed on an OLED screen and sent via Wi-Fi to cloud storage. Doctors can monitor it through mobile apps, and alerts are generated if abnormal IOP is detected.

3.3 PROJECT DESCRIPTION

3.3.1 Overview of the Proposed System

The proposed system is an IoT-enabled smart tonometer designed to measure intraocular pressure for early detection of glaucoma. The device uses a non-contact sensing mechanism based on infrared optical sensing to detect corneal deformation when a controlled stimulus is applied to the eye. The collected sensor data is processed by an embedded AI-enabled controller with Wi-Fi connectivity, which analyses the corneal deformation pattern to estimate the intraocular pressure. The system also includes signal conditioning, noise filtering, display, and power management modules to ensure accurate and reliable measurements. The results are displayed on an OLED screen and can be transmitted through wireless communication for remote monitoring.

3.3.2 System Modules Description

1. IR Rays Driver and Signal Conditioning Circuit

The IR driver and signal conditioning circuit is responsible for generating and controlling the infrared signal used for corneal sensing. This circuit powers the infrared source and ensures stable signal transmission toward the eye. The signal conditioning stage improves the quality of the sensor signal by amplifying weak signals and preparing them for further processing. It ensures that the infrared signal is accurate and stable so that the displacement of the cornea can be measured precisely.

2. IR Optical Displacement Sensor

The IR optical displacement sensor detects the small movement of the cornea when pressure is applied.

3. Embedded AI Controller with Wi-Fi

The embedded controller acts as the central processing unit of the system. It receives filtered sensor data and processes it using embedded algorithms to detect corneal deformation patterns. AI-based processing helps in predicting intraocular pressure more accurately by analysing deformation characteristics. The controller also manages communication between different system modules and supports Wi-Fi connectivity for transmitting data to external devices or cloud platforms for remote monitoring.

4. Corneal Deformation Detection Module

The corneal deformation detection module analyses the displacement data obtained from the optical sensor. When the cornea experiences a slight deformation due to the applied force, the system records the deformation waveform. Important parameters such as deformation amplitude and response time are extracted from the signal.

5. OLED Display Module

The OLED display module provides a visual interface for the user. It shows the measured intraocular pressure values and other system information such as device status and measurement progress. OLED displays are used because they are compact, energy-efficient, and capable of providing clear and bright visuals, making them suitable for portable medical devices.

6. Control Key

The control key allows the user to interact with the device. It is used to start the measurement process, reset the system, or navigate through device functions. This simple interface ensures that the device can be easily operated by medical staff or patients.

7. Power Supply Unit (Rechargeable Unit and Li-Po Battery)

The power supply section consists of a rechargeable unit and a lithium-polymer battery. The rechargeable unit allows the device to be charged when required, while the Li-Po battery provides stable power to all system components.

8. Overall Working Principle

Initially, the infrared driver activates the IR optical sensor system. The sensor measures corneal displacement when the cornea experiences deformation. The signal conditioning and noise filter modules clean and stabilize the sensor signal. The processed data is then sent to the embedded AI controller, where corneal deformation parameters are used to estimate intraocular pressure. The result is displayed on the OLED screen and can also be transmitted through Wi-Fi for remote monitoring. The rechargeable power unit ensures continuous operation of the device. The system also stores patient data for long-term analysis. Additionally, the system is designed to ensure high reliability and user comfort during operation. The signal conditioning unit not only removes noise but also enhances weak signals, ensuring accurate measurements even under different environmental conditions. The embedded AI controller continuously analyses incoming data and improves prediction accuracy by learning from previous measurements. The rechargeable battery is optimized for low power consumption, providing longer operational life and making the device portable and convenient for regular use.

3.4 ADVANTAGES OF PROPOSED SYSTEM

1. Low-Cost and Affordable Design

The proposed system is designed to be cost-effective compared to conventional tonometers. Traditional devices are expensive and require specialized equipment.

2. Compact and Portable Device

Unlike bulky clinical tonometers, the proposed system is compact and lightweight, making it easy to carry and use in different environments.

3. AI-Based Accuracy Improvement

The integration of Artificial Intelligence (AI) in the system improves the accuracy of intra-ocular pressure measurement.

4. Continuous and Real-Time Monitoring

The system supports continuous and real-time monitoring of intra-ocular pressure. Unlike conventional systems that provide only a single measurement, the proposed device can track pressure variations over time.

5. IoT-Based Remote Monitoring

With IoT connectivity, the measured data can be transmitted to a cloud platform. Doctors and healthcare professionals can remotely access the patient's eye pressure data.

6. Non-Invasive and User-Friendly Operation

The device is designed to be non-invasive and easy to operate. It allows patients to perform measurements without the need for medical assistance.

3.5 APPLICATIONS

The IoT-based smart tonometer offers versatile applications across multiple healthcare settings, making it a valuable tool in modern eye care. One of its most significant uses is in home-based glaucoma monitoring, where patients can conveniently measure their intraocular pressure on a regular basis without the need for frequent hospital visits. This not only saves time and cost but also encourages consistent monitoring, which is essential for effective disease management. In remote and rural areas, where access to specialized ophthalmic care is often limited, the device plays a crucial role by enabling basic eye assessments and transmitting data to healthcare professionals located elsewhere. Through tele-ophthalmology platforms, doctors can remotely monitor patients, analyse trends, and provide timely medical advice or intervention when abnormal readings are detected.

Hospitals and clinics can effectively integrate smart tonometers into post-treatment care to enable continuous follow-up and long-term monitoring of intraocular pressure. Unlike occasional in-clinic measurements, these devices provide consistent data over time, allowing clinicians to observe trends and detect subtle fluctuations that may indicate changes in a patient's condition. This ongoing data collection is valuable for assessing the effectiveness of prescribed treatments and making timely, evidence-based adjustments to therapy when necessary. By tracking IOP variations across different times and conditions, healthcare providers can gain deeper insights into individual patient responses and the progression of diseases such as glaucoma. This personalized understanding improves treatment accuracy and patient outcomes. Such data enhances the overall understanding of glaucoma and supports the development of more advanced strategies for diagnosis, monitoring, and long-term ocular health management.

CHAPTER – 4

TECHNICAL SPECIFICATIONS

4.1 HARDWARE REQUIREMENTS

The proposed IoT-based smart tonometer uses multiple hardware modules to measure intraocular pressure accurately. An air-puff driver induces corneal deformation, while an infrared optical sensor detects displacement. Signal conditioning and noise filtering improve data quality. The ESP32 microcontroller processes data using AI, displays results on an OLED screen, and enables communication through UART/Wi-Fi connectivity.

4.1.1 PRIMARY COMPONENTS

i) Spectral / Optical Sensor

The AS7265x spectral sensor is a multi-channel optical sensing module designed for precise light measurement. It operates at 3.3V or 5V and provides 18 spectral channels covering ultraviolet (UV), visible, and infrared (IR) regions. The sensor uses an I2C communication interface, making it easy to integrate with microcontrollers. It has high sensitivity to variations in light intensity and includes built-in optical filters for accurate spectral detection. In this system, it is used to detect corneal deformation by analyzing changes in reflected light.

ii) Air-Puff / Rebound Mechanism

The air-puff or rebound mechanism consists of a micro air pump and a solenoid valve, which work together to generate and control a gentle burst of air. The pump operates at 5V or 12V depending on the model, and the solenoid valve ensures precise air release. This mechanism is controlled using a MOSFET driver circuit and provides fast response time for accurate puffing generation. It is used to direct a controlled air puff onto the cornea to induce deformation for measurement.

iii) Signal Conditioning Circuit

A controlled air puff onto the cornea to induce deformation for measurement. The signal conditioning circuit is built using the LM358 operational amplifier IC, which operates within a voltage range of 3V to 32V.

This circuit is responsible for amplifying and filtering weak analog signals obtained from the optical sensor. It includes a low-pass filter to eliminate high-frequency noise and improve the overall signal-to-noise ratio. The conditioned signal is then sent to the ESP32 ADC for further processing.

iv) OLED Display Module

The OLED display module, based on the SSD1306 controller, is used to display system outputs. It typically has a 0.96-inch screen with a resolution of 128×64 pixels and supports I2C or SPI communication. Operating at 3.3V or 5V, it offers low power consumption and high contrast. In this system, it displays intraocular pressure readings and overall system status clearly to the user.

v) Pressure Sensor

The MPX5010 pressure sensor is used to measure the air pressure generated by the air-puff system. It operates at 5V and provides an analog voltage output corresponding to the pressure range of 0 to 10 kPa. The sensor is known for its high accuracy and stability, making it suitable for safety monitoring. It ensures that the air pressure applied to the eye remains within safe limits.

4.1.2 SECONDARY COMPONENTS

i) External ADC Module

The ADS1115 external ADC module is a 16-bit high-precision analog-to-digital converter that enhances measurement accuracy. It communicates via the I2C interface and includes a programmable gain amplifier for improved signal resolution. This module provides more accurate analog signal conversion compared to the built-in ADC of the ESP32.

ii) Power Supply Module

The power supply module consists of a rechargeable 18650 lithium-ion battery and an AMS1117 voltage regulator. It provides stable output voltages of 3.3V and 5V required for different components in the system. The module also includes protection features such as over-voltage and short-circuit protection, ensuring reliable and safe operation.

iii Wi-Fi / UART Communication

The communication system uses the built-in Wi-Fi capability of the ESP32 along with UART support for serial communication. It enables reliable and efficient data exchange between the device and external systems. The system also supports secure communication protocols, ensuring safe transmission of sensitive medical data. Through Wi-Fi connectivity, it can link to cloud platforms or IoT dashboards, allowing intraocular pressure data to be transmitted for real-time remote monitoring, visualization, and analysis by healthcare professionals.

iv) User Interface Components

The user interface components of the system include push buttons or touch sensors for user input, a buzzer for alert notifications, and status LEDs for visual indication. These elements allow users to interact easily with the device and operate it efficiently. They also provide real-time feedback on the system's functioning, such as measurement status, alerts, or errors. This improves usability, ensures quick response to critical conditions, and makes the overall device more intuitive, reliable, and user-friendly for continuous monitoring applications.

4.2 SOFTWARE REQUIREMENTS

The software requirements of the proposed IoT-enabled smart tonometer system are essential for enabling efficient data acquisition, processing, analysis, and communication between hardware components and cloud platforms. The system combines embedded programming, Internet of Things frameworks, and machine learning techniques to achieve seamless functionality. It supports continuous intraocular pressure monitoring with real-time data transmission and storage. Additionally, advanced Machine Learning algorithms, particularly a Convolutional Neural Network, are implemented for accurate glaucoma detection and classification. This integration ensures reliable performance, real-time decision-making, and improved diagnostic accuracy, making the system effective for modern healthcare applications and remote patient monitoring.

4.2.1 PRIMARY COMPONENTS

i) Embedded C

Embedded C is a low-level programming language widely used for programming

microcontrollers and embedded systems. It allows direct interaction with hardware components such as GPIO, ADC, UART, and I2C interfaces, enabling precise and efficient control of electronic systems. Embedded C is well-suited for real-time applications due to its fast execution speed, low memory consumption, and support for interrupt handling. In this project, Embedded C is used to program the ESP32 for sensor data acquisition, signal processing, and overall system control, ensuring reliable and efficient operation of the embedded IoT system.

ii) Python

Python is a high-level, versatile programming language known for its simplicity and readability. It is widely used in fields such as machine learning, artificial intelligence, data science, and automation due to its extensive library support. Python provides powerful tools for data processing, statistical analysis, and visualization, making it highly effective for developing intelligent systems. It also offers cross-platform compatibility and enables rapid prototyping, which speeds up development cycles. In this system, Python is used for AI model development, data analysis, and cloud-side processing of intraocular pressure data, ensuring efficient handling and meaningful interpretation of collected sensor information.

iii) AI / Machine Learning Model

The AI/Machine Learning model used in this system is based on a Convolutional Neural Network (CNN) algorithm. It is capable of automatically extracting important features from sensor data and provides high accuracy in pattern recognition. The model effectively filters noise caused by blinking and motion artifacts and also performs sensor drift compensation. It supports real-time prediction, enabling fast and reliable analysis. In this project, the CNN model is used to analyze corneal deformation patterns accurately for improved intraocular pressure measurement.

4.2.2 SECONDARY COMPONENTS

i) Thonny IDE

Thonny IDE is a simple and lightweight Python development environment designed for ease of use, especially for beginners and educational purposes. It comes with a built-in Python interpreter, eliminating the need for complex setup, and offers features

such as step-by-step debugging, variable inspection, and a clean, user-friendly interface. These capabilities make it easier to understand program execution and identify errors. In this project, Thonny IDE is used for developing, testing, and refining AI models efficiently. Its Simplicity allows quick prototyping and experimentation, while its debugging tools help ensure accuracy and reliability in model implementation.

ii) Visual Studio Code

Visual Studio Code is a powerful and versatile code editor that supports multiple programming languages, including Embedded C and Python. It provides integrated debugging tools, intelligent code completion, and a rich ecosystem of extensions that enhance development efficiency. Additionally, its built-in Git integration allows seamless version control, making it easier to track changes and collaborate on projects. In this system, Visual Studio Code is used for writing, debugging, and managing code across different modules. Its flexibility and robust features make it highly suitable for handling complex project development and ensuring smooth integration of all system components.

iii) Thing Speak IoT Cloud Platform

ThingSpeak is a cloud-based IoT platform designed for data storage, visualization, and analysis. It enables real-time monitoring through interactive dashboards, which helps in performing advanced data processing and extracting meaningful insights. ThingSpeak integrates seamlessly with microcontrollers like ESP32, enabling efficient data transmission and remote access for both doctors and users. It also provides features such as alerts, notifications, and scalable architecture for handling growing datasets. In this project, ThingSpeak is used to store intraocular pressure data and support remote monitoring and analysis.

iv) Communication Protocol

Communication protocols such as MQTT and HTTP are widely used for efficient data transmission in IoT-based systems. These lightweight protocols enable real-time communication between devices and cloud platforms, ensuring fast and reliable data exchange. They also support secure communication through SSL/TLS encryption, protecting sensitive information during transmission. Their low bandwidth consumption and high efficiency make them ideal for resource-constrained devices like ESP32.

CHAPTER 5

SOFTWARE DESCRIPTION

5.1 OVERVIEW OF SOFTWARE ARCHITECTURE

The software architecture of the IoT Smart Tonometer for AI-Based Intra-Ocular Pressure Prediction and Early Glaucoma Detection is designed as a layered and modular framework to ensure reliable data acquisition, accurate pressure estimation, secure communication, and user-friendly visualization. The architecture integrates embedded firmware, signal processing algorithms, artificial intelligence models, and cloud-based services to enable real-time monitoring and remote healthcare support. At the lowest level, the embedded software running on the ESP32 microcontroller handles direct interaction with the hardware components. This layer is responsible for controlling the spectral imaging sensor, managing timing and synchronization, acquiring raw optical signals related to corneal deformation, and performing initial data validation. The firmware is developed in Embedded C to ensure low latency, efficient memory usage, and real-time performance. The ESP32 also manages peripheral devices such as the OLED display and communication interfaces.

The next layer consists of the signal processing and preprocessing module, where raw sensor data is filtered to remove noise caused by blinking, minor head movements, and ambient light interference. This layer normalizes and formats the processed data to make it suitable for AI-based analysis. By handling preprocessing at the device level, the system reduces unnecessary data transmission and improves prediction accuracy. Above this lies the AI-based prediction layer, implemented using Python-based machine learning models. This layer extracts meaningful features from the processed spectral data and estimates intra-ocular pressure values. The modular design allows the model to be updated or retrained without modifying the embedded firmware. The IoT communication layer enables seamless data transmission between the ESP32 device and the cloud platform using Wi-Fi connectivity. This layer ensures secure data transfer through lightweight communication protocols and supports real-time synchronization. It acts as a bridge between the embedded system and higher-level applications.

The cloud and application layer manages long-term data storage, trend analysis, and user interaction. Processed IOP data is stored in a centralized database, allowing doctors to monitor patient history remotely. The mobile or web-based application provides visual dashboards, alert notifications, and report generation features for both patients and healthcare professionals. Overall, the software architecture is designed to be scalable, secure, and reliable, supporting continuous monitoring, AI-driven analysis, and remote medical supervision. The clear separation of responsibilities across layers ensures ease of maintenance, flexibility for future enhancements, and efficient integration of advanced technologies for early glaucoma detection.

5.2 EMBEDDED FIRMWARE DESIGN (ESP32)

The embedded firmware forms the core of the smart tonometer system and is implemented on the ESP32 microcontroller. The firmware is responsible for coordinating all low-level operations, including sensor control, data acquisition, preprocessing coordination, display updates, and wireless communication. Embedded C is used for firmware development to ensure efficient memory usage, fast execution, and real-time responsiveness. The ESP32 firmware initializes all hardware peripherals at system startup, including the spectral imaging sensor interface, OLED display, and Wi-Fi module. Timers and interrupts are configured to ensure accurate sampling of sensor data during the corneal deformation measurement process.

The firmware continuously monitors system status and ensures synchronized data capture during the air-puff or rebound event. In addition, the ESP32 manages wireless communication by packaging processed sensor data and transmitting it securely to the cloud platform. Error handling routines include managing communication failures, sensor disconnections, and abnormal operating conditions. The embedded firmware thus acts as a reliable control layer that ensures smooth interaction between hardware components and higher-level AI and cloud services.

Introduction to Embedded ‘C’

Embedded systems programming is different from developing applications on a desktop computer. Key characteristics of an embedded system, when compared to PCs, are as follows:

- Embedded devices have resource constraints (limited ROM, limited RAM, limited stack space, less processing power)
- Components used in embedded systems and PCs are different; embedded systems typically use smaller, less power consuming components.
- Two salient features of Embedded Programming are code speed and code size. Code speed is governed by processing power, timing constraints, whereas code size is governed by available program memory and use of programming language.
- Goal of embedded system programming is to get maximum features in minimum space and minimum time.

Embedded systems are programmed using different types of languages:

- Machine Code
- Low level language, i.e., assembly
- High level language like C, C++, Java, Ada, etc.
- Application-level languages like Visual Basic, scripts, Access, etc...

Assembly language seems to be an obvious choice for programming embedded devices. However, use of assembly language is restricted to developing efficient codes in terms of size and speed. Also, assembly codes lead to higher software development costs and code portability is not there. Developing small codes are not much of a problem, but large programs/projects become increasingly difficult to manage in assembly language. Finding good assembly programmers has also become difficult nowadays. Hence high-level languages are preferred for embedded systems programming.

5.3 PROTEUS

Proteus was initially created as a multiplatform (DOS, Windows, Unix) system utility, to manipulate text and binary files and to create CGI scripts. The language was later focused on Windows, by adding hundreds of specialized functions for: network and serial communication, database interrogation, system service creation, console applications, keyboard emulation, ISAPI scripting (for IIS). Most of the additional functions in Proteus are primarily available in the Windows version of the interpreter, although a Linux version is also supported. The language was designed with a strong focus on practicality, ensuring it is easy to use, efficient, complete, readable, and consistent in structure and execution.

Proteus also provides a wide range of advanced data structures, including arrays, single and double-ended queues, stacks, bitmaps, sets, and AVL trees, which enhance its ability to manage complex data efficiently. Additionally, the language is highly extensible, allowing users to add custom functions written directly in Proteus or integrate external libraries through DLLs created in C or C++, making it flexible for advanced applications.

5.4 ARDUINO C SOFTWARE INSTALLATION

Follow the steps to Arduino c installation.

Step 1: Visit <http://www.arduino.cc/en/main/software> to download the latest Arduino.

Step 2: Save the .exe file to your hard drive.

Step 3: Open the .exe file.

Step 4: Click the button to agree to the licensing agreement.



Figure 5.1 License Agreement

Step 5: Decide which components to install, then click “Next”.

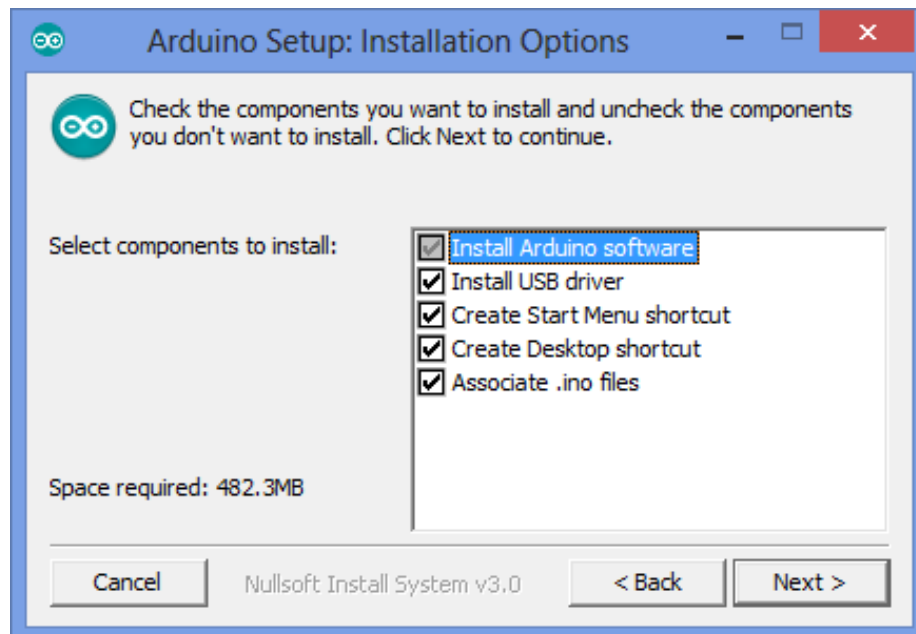


Figure 5.2 Installation option

Step 6: Select which folder to install the program to, then click “Install”.

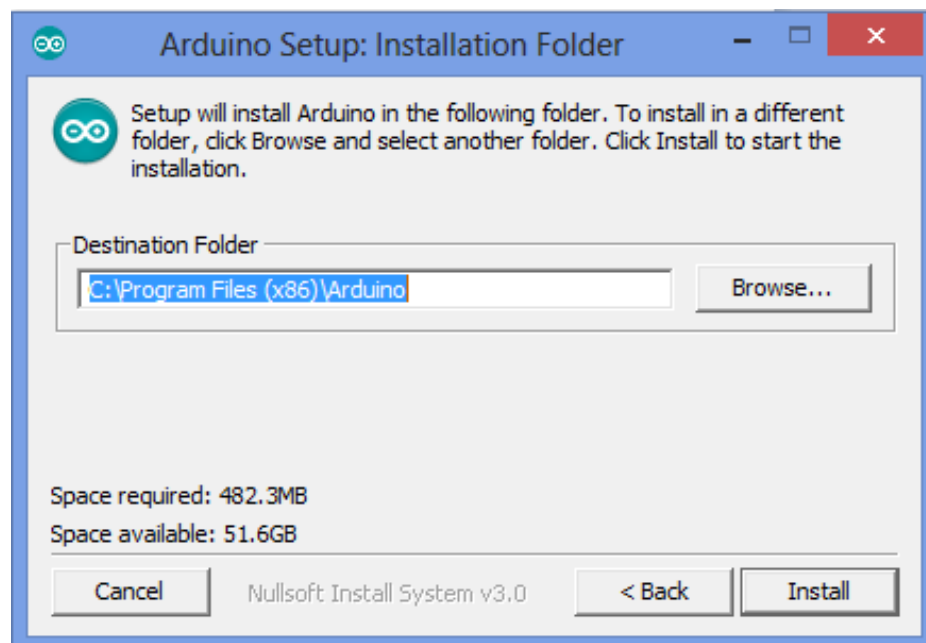


Figure 5.3 Installation folder

Step 7: Wait for the program to finish installing, then click “Close”.

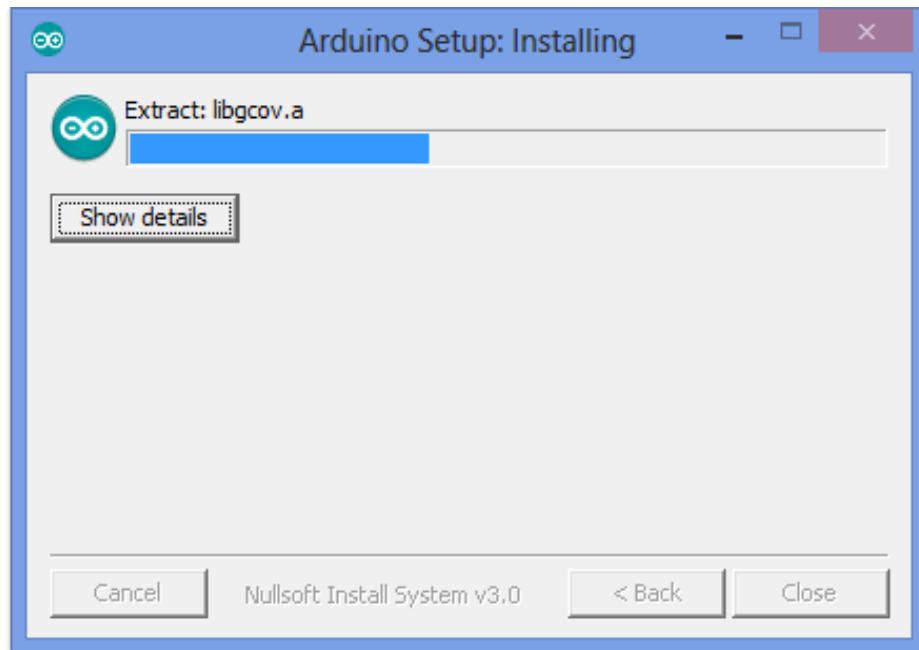


Figure 5.4 Installing process

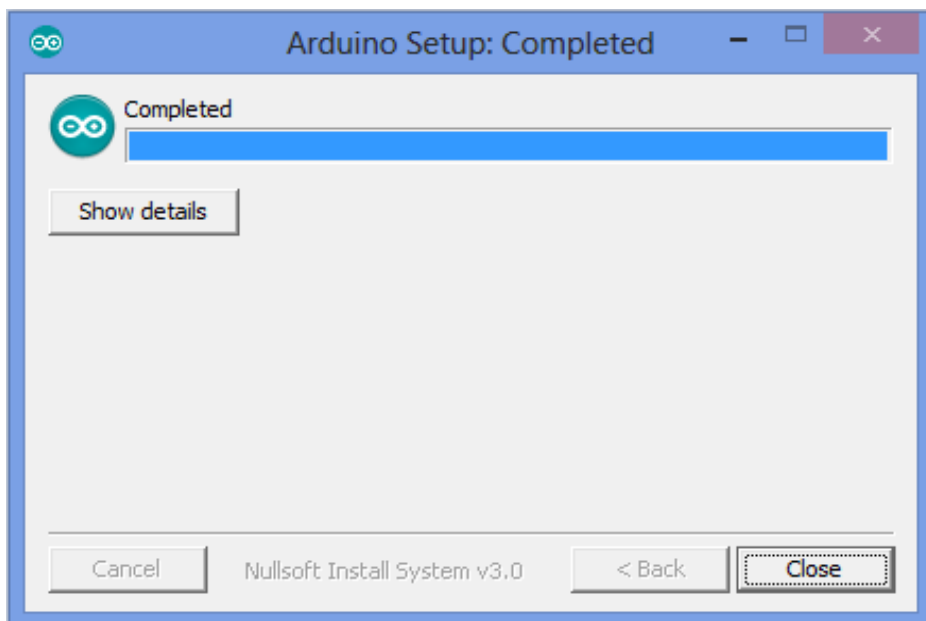


Figure 5.5 Process completion

Step 8: Now find the Arduino shortcut on your Desktop and click on it.

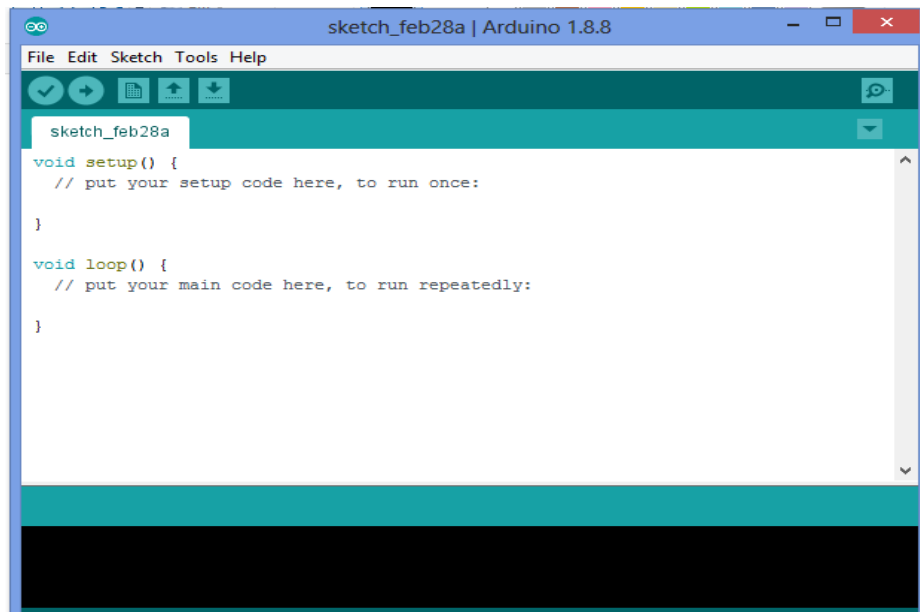


Figure 5.6 Arduino 1.8.8

Step 9: Open new file

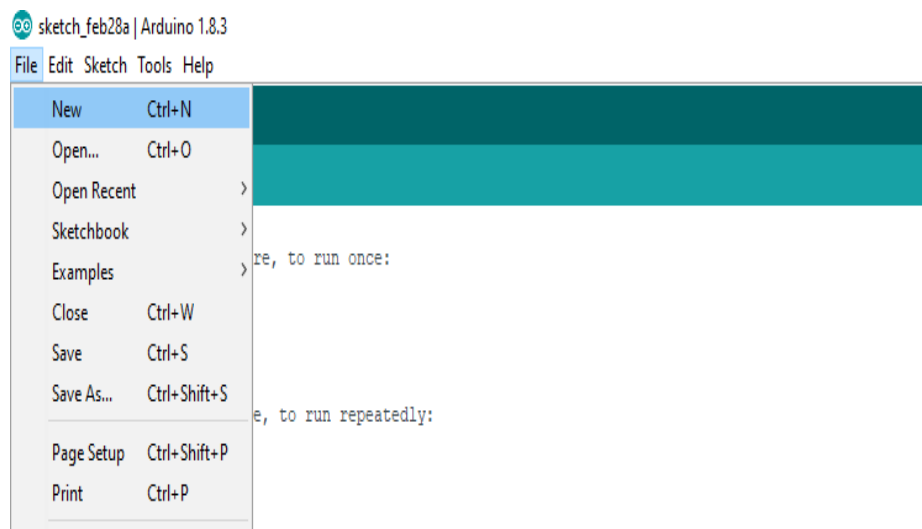


Figure 5.7 New File Creation

5.5 SPECTRAL IMAGING SENSOR DATA ACQUISITION

The spectral imaging sensor data acquisition software is a crucial component of the smart tonometer system, responsible for capturing high-resolution optical data that reflects corneal deformation during intra-ocular pressure measurement. Unlike conventional optical sensing systems that rely on single-wavelength intensity measurements, the spectral imaging sensor records wavelength-dependent responses. This enables the system to obtain detailed and multi-dimensional information about corneal behaviour when subjected to a controlled air-puff or rebound force, thereby improving the accuracy and depth of analysis. The ESP32 microcontroller communicates with the spectral imaging sensor through a dedicated digital interface such as SPI or I2C. The firmware is designed to configure and control various sensor parameters, including sampling rate, exposure time, resolution, and wavelength range.

Proper configuration of these parameters is essential to ensure that the captured data is both precise and consistent under different operating conditions. The software dynamically adjusts these settings if required, based on environmental conditions or system requirements, ensuring optimal performance during each measurement cycle. During operation, the sensor captures a sequence of spectral frames that represent the deformation pattern of the cornea over time. These frames provide a time-dependent profile of corneal displacement, which is critical for accurate IOP estimation. The acquired data is temporarily stored in buffer memory within the ESP32 to maintain synchronization and prevent data loss. Efficient buffering ensures that continuous data streams are handled smoothly without interruption, even during rapid acquisition cycles.

Each data frame is time-stamped to maintain proper sequencing and to allow precise temporal analysis. The software also includes validation mechanisms to ensure data integrity. It checks for corrupted, incomplete, or inconsistent data samples, which may occur due to communication errors or external disturbances. Such invalid data is automatically discarded or corrected to prevent inaccuracies in subsequent processing stages. Overall, the spectral imaging sensor data acquisition software ensures precise, stable, and repeatable data collection. Its robust design enhances system reliability and plays a key role in enabling accurate glaucoma detection and continuous eye pressure monitoring in both clinical and home-based environments.

5.6 SIGNAL PROCESSING AND NOISE FILTERING ALGORITHMS

Signal processing plays a vital role in enhancing the accuracy and reliability of intra-ocular pressure estimation in the proposed smart tonometer system. The raw data acquired from the spectral imaging sensor often contains various types of noise and disturbances that can affect measurement quality. These disturbances may arise due to involuntary blinking, slight head movements, variations in ambient lighting, and inherent sensor drift over time. If left unprocessed, such noise can distort the true corneal deformation signal and lead to inaccurate IOP predictions. Therefore, an effective signal processing stage is essential to refine the data before it is used for further analysis.

To address these challenges, the system incorporates dedicated preprocessing and noise filtering algorithms. Initially, smoothing filters such as moving average or low-pass filters are applied to reduce high-frequency noise components while preserving the essential characteristics of the signal. Normalization techniques are then used to scale the data within a consistent range, ensuring uniformity across different measurements and users. Baseline correction methods help remove any constant offsets or drift in the signal, stabilizing the data and making it more suitable for analysis. In addition to these techniques, temporal filtering is employed to eliminate sudden spikes or irregularities caused by blinking or abrupt eye movements. This ensures that only relevant and stable portions of the signal are considered for IOP estimation.

Sensor drift compensation methods are also implemented to correct gradual changes in sensor output that may occur due to temperature variations or prolonged usage. These techniques help maintain long-term accuracy and consistency of the measurements. Once the noise is effectively reduced and the signal is refined, the processed data represents a more accurate depiction of corneal deformation. Overall, the signal processing and noise filtering stage significantly enhances the robustness of the system. It minimizes the impact of environmental and physiological disturbances, reduces the likelihood of false readings, and ensures consistent performance under real-world conditions. This makes the smart tonometer more reliable for continuous monitoring and early detection of glaucoma, especially in non-clinical and home-based environments.

5.7 AI-BASED INTRA-OCULAR PRESSURE PREDICTION MODEL

The AI-based intra-ocular pressure prediction model is a core component of the smart tonometer system, responsible for transforming processed spectral sensor data into accurate and reliable IOP values. The relationship between corneal deformation and intra-ocular pressure is highly nonlinear and influenced by multiple physiological and environmental factors. Because of this complexity, traditional mathematical or linear models are not sufficient to achieve precise estimation. To overcome these limitations, an artificial intelligence-based approach is adopted, enabling the system to learn complex patterns and improve prediction accuracy.

The AI model is developed using Python and is trained on carefully extracted features derived from spectral imaging sensor data. These features include deformation intensity, wavelength-dependent response patterns, temporal variations, and normalized signal amplitudes. Each of these parameters reflects important physical characteristics of corneal behaviour when subjected to a controlled force. By selecting meaningful and representative features, the model is able to focus on the most relevant aspects of the data, improving its predictive capability.

During the training phase, the model is fed with labelled datasets that consist of sensor-derived features along with corresponding reference IOP values obtained from standard clinical instruments. This allows the AI system to learn the complex mapping between input features and actual pressure values. Advanced learning techniques, such as Convolutional Neural Networks or regression-based algorithms, may be used to capture both spatial and temporal relationships within the data. The model parameters are optimized through iterative training processes to minimize prediction error and improve accuracy.

Once trained, the AI model is deployed within the system for real-time operation. During measurement, the processed spectral features are input into the model, which then estimates the corresponding intra-ocular pressure. The model is designed to handle variations caused by differences in corneal properties among individuals, as well as external factors such as sensor drift or environmental changes. This adaptability ensures consistent and reliable performance across different users and conditions.

5.8 COMMUNICATION PROTOCOLS AND IOT INTEGRATION

The IoT integration layer plays a vital role in enabling seamless communication between the smart tonometer device and external platforms for remote monitoring, data storage, and analysis. At the core of this layer is the ESP32 microcontroller, which features built-in wi-fi capability and functions as the communication gateway of the system. This integration eliminates the need for additional communication modules, thereby reducing system complexity, cost, and size while maintaining efficient connectivity. Once the intra-ocular pressure values are predicted by the ai model, the ESP32 organizes the data into structured formats such as packet-based frames. These data packets may include IOP readings, timestamps, device status, and alert indicators. The formatted data is then transmitted securely to a cloud platform using standard internet protocols such as http or MQTT. These protocols are widely used in IoT systems due to their efficiency, reliability, and low bandwidth requirements.

To ensure consistent and reliable communication, especially in environments with unstable network connectivity, the system incorporates mechanisms such as automatic reconnection, retry attempts, and connection status monitoring. If a transmission fails, the esp32 temporarily stores the data in local memory and resends it once the connection is restored. This prevents data loss and ensures continuous monitoring without interruption. Security is a critical aspect of IoT-based healthcare systems. The proposed design implements secure communication practices such as data encryption and authentication protocols to protect sensitive patient information during transmission. This ensures that only authorized users and healthcare providers can access the data, maintaining privacy and compliance with healthcare standards.

Through IoT integration, real-time IOP readings, historical trends, and alert notifications are made accessible to doctors and caregivers via mobile or web applications. Overall, the IoT-enabled architecture enhances the functionality, accessibility, and efficiency of the smart tonometer. It supports continuous patient monitoring outside clinical environments, reduces the need for frequent hospital visits, and promotes proactive healthcare management. This makes the system highly suitable for home-based monitoring, tele-ophthalmology applications, and modern digital healthcare solutions.

5.9 CLOUD PLATFORM AND DATA MANAGEMENT

The cloud platform plays a vital role in enabling remote monitoring, long-term storage, and advanced analysis of Intra-Ocular Pressure data in the proposed smart tonometer system. Once the ESP32 microcontroller predicts the IOP value using the integrated AI model, the data is securely transmitted to the cloud through the IoT communication layer. The cloud infrastructure is designed to efficiently handle continuous data streams from multiple devices, ensuring high availability, scalability, and reliability. This makes it suitable for real-time healthcare applications where uninterrupted data access is critical. On the cloud side, incoming data packets are first validated to ensure accuracy and completeness. This includes checking for missing values, incorrect formats, or transmission errors.

Once validated, the data is stored in a structured database along with essential metadata such as timestamps, device identifiers, and patient IDs. This organized storage mechanism allows the system to maintain comprehensive historical records for each user. These records are essential for tracking long-term trends in intra-ocular pressure, which is a key factor in the early detection and management of glaucoma. The cloud platform also performs data aggregation and trend analysis by processing stored pressure values over time. These patterns can indicate the onset or progression of glaucoma or other ocular conditions. By continuously analyzing incoming data, the system can generate alerts and notifications when abnormal trends are detected, enabling timely medical intervention.

In addition, the cloud-based system provides a user-friendly interface through web or mobile applications, allowing healthcare professionals to remotely access patient data. Doctors can view real-time readings, compare current values with historical data, and assess the effectiveness of treatment plans. This remote accessibility improves clinical decision-making and reduces the need for frequent in-person visits. The centralized nature of the cloud platform ensures easy system maintenance, data backup, and scalability. It also allows integration with other healthcare systems such as electronic health records (EHRs) and telemedicine platforms. Advanced security measures, including encryption and authentication protocols, are implemented to protect sensitive patient data.

5.10 MOBILE / WEB APPLICATION SOFTWARE

The mobile and web application software provides an intuitive user interface for patients and doctors to interact with the smart tonometer system. The application retrieves real-time and historical IOP data from the cloud and presents it in a clear, graphical format. Users can view current pressure values, daily and weekly trends, and alert notifications through dashboards designed for easy interpretation. For patients, the application supports simple visualization of their eye pressure status, encouraging regular self-monitoring and adherence to treatment plans. For doctors, advanced views are provided to analyse long-term trends, compare multiple readings, and assess risk levels. The application also supports role-based access control, ensuring that sensitive medical data is accessible only to authorized users. Additionally, the mobile/web application enables report generation and data export for medical documentation and follow-up consultations. By providing remote access to patient data, the application reduces the need for frequent hospital visits and enhances the effectiveness of tele-ophthalmology services.

5.11 ALERT AND NOTIFICATION SYSTEM

The alert and notification system is designed to provide timely warnings to patients and healthcare professionals when abnormal intra-ocular pressure levels are detected. Once the AI-based prediction model estimates the IOP value, the system compares it with predefined safe and critical threshold limits.

If the measured pressure exceeds the normal range or shows a sudden rise compared to historical trends, an alert is triggered automatically. The ESP32 communicates alert information to the cloud platform, where notification logic is executed. Alerts can be delivered through the mobile or web application in the form of visual warnings, messages, or notifications.

This immediate feedback enables patients to seek medical attention at an early stage and allows doctors to take proactive measures. The alert system plays a crucial role in preventing delayed diagnosis and supports continuous glaucoma monitoring.

5.12 SECURITY AND DATA PRIVACY IMPLEMENTATION

Security and data privacy are essential considerations in the smart tonometer system due to the sensitive nature of medical information. The software architecture incorporates multiple layers of protection to ensure secure data handling and prevent unauthorized access. During data transmission from the ESP32 to the cloud platform, secure communication protocols are used to protect patient information from interception.

Access to cloud-stored data and mobile or web applications is controlled through authentication and authorization mechanisms. Only authorized users such as registered patients and doctors are permitted to view or modify data. Basic encryption techniques are applied to stored data to safeguard patient privacy. These security measures ensure compliance with healthcare data protection requirements and maintain user trust in the system.

5.13 OLED DISPLAY CONTROL SOFTWARE

The OLED display control software is an essential part of the smart tonometer system, responsible for presenting real-time intra-ocular pressure (IOP) readings and system status directly on the device. It acts as the primary user interface, allowing users to interact with the system and instantly view measurement results without requiring external devices. The ESP32 microcontroller manages the display operations by interfacing with the OLED module through communication protocols such as I2C or SPI, ensuring fast and reliable data transfer. It continuously retrieves processed data from the ESP32, including the predicted IOP values, and renders it in a clear and structured format.

The display typically shows key parameters such as current IOP readings, device status, battery level, wi-fi connectivity status, and measurement progress indicators. This comprehensive information helps users understand the system's operation and verify that the measurement process is functioning correctly. Furthermore, the OLED display provides standalone functionality, allowing the device to operate independently even when network connectivity is unavailable. Users can still view their IOP readings and basic system information directly on the device without relying on cloud platforms or mobile applications.

5.14 SOFTWARE DEVELOPMENT TOOLS AND IDE

The software development for the IoT-based smart tonometer is carried out using reliable, efficient, and scalable programming tools and development environments to ensure accurate performance and seamless integration of all system components. The system combines embedded programming, artificial intelligence, and cloud computing technologies to achieve real-time intraocular pressure monitoring and early glaucoma detection. Overall, these software tools and technologies work together to enable efficient development, testing, debugging, and maintenance of the IoT smart tonometer system. They ensure reliable seamless integration of embedded systems, artificial intelligence, and IoT components.

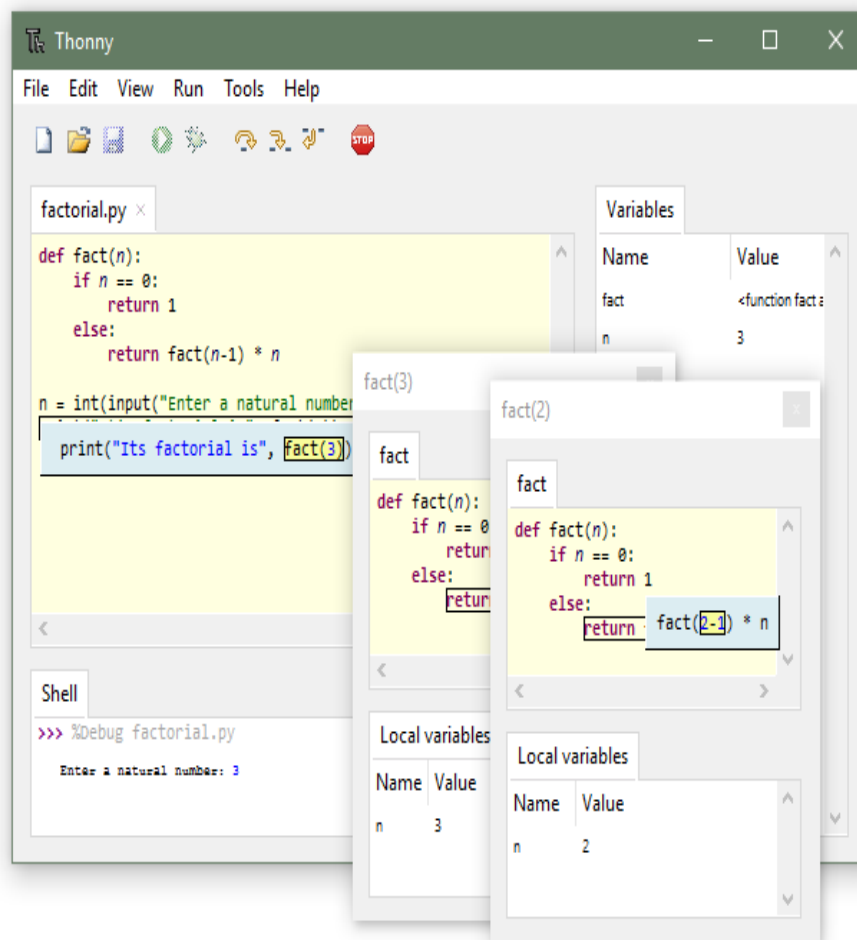


Figure 5.8 Thonny IDE Interface

VS CODE

Visual Studio Code is a free and open-source code editor developed by Microsoft. It supports multiple programming languages such as Python, C++, and embedded C, and provides a lightweight yet powerful environment for coding and debugging. In the proposed smart tonometer system, VS Code is used as the primary development platform for programming the ESP32 microcontroller. It allows developers to write, edit, and debug the embedded code efficiently. Extensions related to IoT and microcontroller programming help in uploading code, monitoring serial output, and troubleshooting errors. Its integrated terminal and debugging features improve development speed and accuracy. Thus, VS Code plays an important role in implementing and testing the software part of the smart tonometer system.

Additionally, VS Code provides features like IntelliSense, which offers smart code suggestions and auto-completion, reducing coding errors and improving productivity. The built-in debugging tools help in identifying logical and runtime errors effectively by allowing step-by-step execution and variable monitoring. This is especially useful while developing AI-based and sensor-interfacing modules in projects. VS Code also supports version control integration using Git, which helps in tracking code changes, managing different versions, and collaborating efficiently.

Visual Studio Code offers a wide range of extensions, such as Platform IO and Arduino support, making it highly suitable for embedded systems and IoT-based projects like the smart tonometer. These extensions simplify coding, debugging, and deployment for microcontroller-based applications. Additionally, its cross-platform compatibility allows it to run smoothly on Windows, Linux, and macOS, ensuring accessibility for a wide range of developers. The integrated terminal, version control, and debugging tools further enhance productivity. Overall, Visual Studio Code provides a flexible, efficient, and user-friendly development environment that significantly improves the process of building, testing, and managing complex embedded and IoT systems.

CHAPTER 6

HARDWARE DESCRIPTION

6.1 OVERVIEW OF HARDWARE ARCHITECTURE

The hardware architecture of the IoT Smart Tonometer for AI-based intra-ocular pressure prediction and early glaucoma detection is designed as a compact, modular, and portable system. It enables accurate sensing, real-time processing, and efficient wireless communication. The architecture integrates multiple modules such as sensing, actuation, signal conditioning, processing, display, and connectivity to support continuous and non-invasive monitoring of intra-ocular pressure, making it suitable for modern healthcare applications.

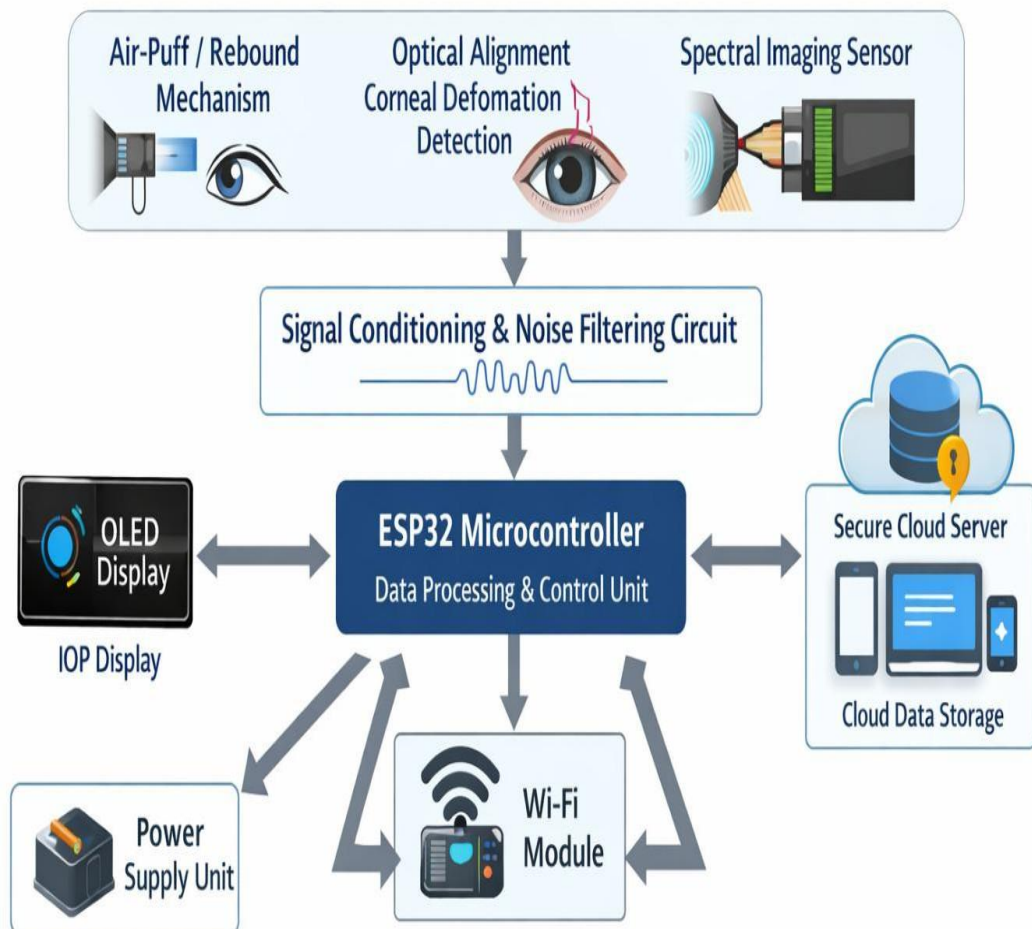


Figure 6.1 Hardware Architecture of IOP Monitoring System

At the sensing layer, a spectral imaging sensor is used to capture detailed optical information related to corneal deformation during a controlled air-puff or rebound-based measurement process. This sensing mechanism ensures a completely non-invasive and safe method for measuring intraocular pressure. The sensor is supported by an optical alignment and deformation detection assembly, which helps maintain correct eye positioning and ensures consistent measurements, thereby improving accuracy and repeatability. The analog signals obtained from the sensor are passed through a signal conditioning and noise filtering circuit, which removes unwanted noise, reduces signal distortion, and stabilizes the output before digital processing.

The ESP32 microcontroller serves as the central control and processing unit of the system. It manages sensor interfacing, timing synchronization, and efficient data acquisition, while also coordinating with ai-based algorithms for intelligent analysis of IOP data. The processed information is displayed locally using an OLED display module, providing real-time feedback such as IOP values, alerts, and system status to the user in a clear and user-friendly manner.

For remote monitoring, the ESP32's built-in wi-fi module enables secure IoT connectivity with cloud platforms, allowing data storage, sharing, and access for further analysis. The cloud system supports long-term patient data tracking and helps detect early signs of glaucoma through trend analysis. A regulated power supply unit ensures stable and reliable operation of all hardware components. Additionally, the system is designed with modularity and scalability in mind, allowing easy integration of new features, software updates, and hardware upgrades. This flexibility enhances system longevity and adaptability.

Furthermore, the system incorporates efficient data handling techniques to ensure minimal latency during real-time monitoring and transmission. Advanced encryption protocols can be implemented to protect sensitive patient data during wireless communication, ensuring privacy and security. The integration of AI models, such as CNNs, enables continuous learning and improvement in prediction accuracy over time. The system can also be extended with mobile application support, allowing users and doctors to access real-time data, receive alerts, and monitor trends conveniently.

6.2 SPECTRAL IMAGING SENSOR

The spectral imaging sensor is a critical hardware component in the proposed IoT-based smart tonometer, as it enables precise and detailed observation of corneal behaviour during intraocular pressure measurement. Unlike conventional optical sensors that capture only light intensity at a single wavelength, spectral imaging sensors acquire information across multiple wavelengths. This multi-wavelength capability provides a richer and more comprehensive dataset, allowing the system to analyse subtle optical variations associated with corneal deformation more effectively.

In this system, the spectral imaging sensor is carefully positioned to monitor the cornea when a controlled air-puff or rebound force is applied. As the cornea deforms in response to this stimulus, the sensor captures changes in reflected or transmitted light across different wavelengths. These changes form unique spectral signatures that are directly influenced by the biomechanical properties of the cornea, such as elasticity, thickness, and stiffness, as well as the internal intraocular pressure. Since higher IOP results in reduced deformation and lower IOP causes greater deformation, the spectral data provides valuable insights into pressure variations.

One of the major advantages of using a spectral imaging sensor is its high sensitivity to minute changes in corneal structure. It can detect even slight variations that may not be captured by traditional single-wavelength sensors. Additionally, the multi-spectral approach helps minimize the effects of external disturbances such as ambient lighting conditions, eye movements, and minor misalignments. This improves the reliability and consistency of measurements, making the system suitable for real-world, non-clinical environments.

The analog output generated by the sensor is first conditioned and filtered to remove noise and unwanted interference. The microcontroller processes this high-dimensional spectral data and forwards it to the AI-based model, such as a Convolutional Neural Network, for accurate IOP prediction. Overall, the spectral imaging sensor forms the foundation of the smart tonometer's sensing mechanism. By providing detailed, high-resolution optical data, it enables advanced signal processing and intelligent analysis, ensuring accurate, reliable, and non-invasive intraocular pressure measurement for early glaucoma detection and continuous monitoring.



Figure 6.2 Spectral Imaging Sensor

6.3 SIGNAL CONDITIONING AND NOISE FILTERING MODULE

The signal conditioning and noise filtering module plays a crucial role in ensuring the accuracy, stability, and reliability of intra-ocular pressure measurements in the proposed smart tonometer system. The raw signals obtained from the spectral imaging sensor are often affected by various sources of noise and interference, such as ambient light fluctuations, involuntary eye movements, blinking, and electronic disturbances from surrounding components. If these unwanted variations are not properly handled, they can distort the actual corneal deformation signal and lead to inaccurate IOP estimation. Therefore, this module acts as an essential preprocessing stage before the data is sent to the ESP32 microcontroller.

The signal conditioning stage involves multiple processes designed to enhance the quality of the sensor output. Initially, amplification circuits are used to increase the strength of weak signals so that subtle corneal deformation patterns can be clearly detected and analysed. This is particularly important because the deformation induced by the air-puff or rebound mechanism is very small and requires precise measurement. Following amplification, filtering techniques are applied to remove unwanted noise components. Low-pass filters are commonly used to eliminate high-frequency noise caused by electronic interference, while maintaining the integrity of the useful signal.

In addition to basic filtering, advanced techniques such as adaptive filtering can be implemented to dynamically adjust to changing noise conditions. This is especially beneficial in real-world environments where factors like slight head movement or blinking can introduce unpredictable disturbances. The module may also include signal stabilization and smoothing circuits to ensure consistent output and reduce sudden fluctuations. These processes help in maintaining a clean and continuous signal waveform that accurately represents corneal deformation. The conditioned signal is then converted into a stable and reliable form suitable for digital processing by the ESP32 microcontroller. By providing high-quality input data, this module significantly enhances the performance of the AI-based prediction model, such as the Convolutional Neural Network, used for IOP estimation. Clean data allows the algorithm to extract meaningful features more effectively, leading to improved prediction accuracy.

6.4 POWER SUPPLY

The power supply unit (PSU) is a fundamental component of the IoT Smart Tonometer, responsible for delivering stable and regulated electrical power to all hardware modules. Since the system is designed for portability and home-based monitoring, the PSU is carefully designed to ensure low power consumption, high efficiency, and reliable performance over extended periods. A stable power source is essential for maintaining the accuracy of sensor readings, proper functioning of the ESP32 microcontroller, and consistent operation of peripheral components such as the OLED display and signal conditioning circuits. In the proposed system, a linear regulated power supply is used to provide a constant 5V output, which is suitable for driving the microcontroller and other electronic components. The PSU typically includes a rechargeable battery, making the device portable and convenient for repeated use without frequent replacement. The voltage regulation circuitry ensures that fluctuations in input voltage do not affect the output, thereby protecting sensitive components from damage and ensuring consistent system performance. The regulated 5V output is distributed to various modules, including the spectral imaging sensor, signal conditioning unit, and ESP32.

For example, a 5V regulated supply,

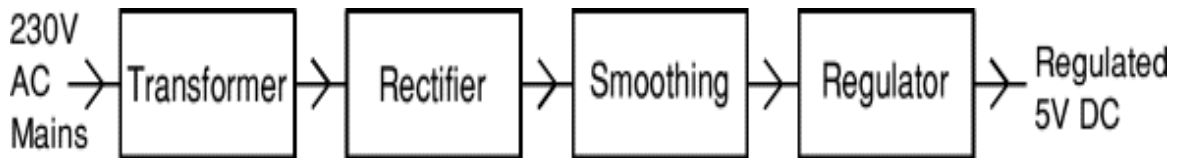
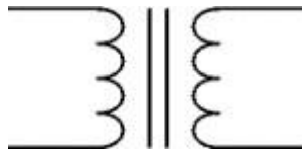


Figure 6.3 Power Supply System



Transformer Symbol

Transformers convert AC electricity from one voltage to another with little loss of power. Transformers work only with AC, and this is one of the reasons why mains electricity is AC. Step-up transformers increase voltage, step-down transformers reduce voltage. Most power supplies use a step-down transformer to reduce the dangerously high mains voltage (230V in UK) to a safer low voltage. The input coil is called the primary and the output coil is called the secondary. There is no electrical connection between the two coils; instead, they are linked by an alternating magnetic field created in the soft-iron core of the transformer.

The two lines in the middle of the circuit symbol represent the core. Transformers waste very little power so the power out is (almost) equal to the power in. Note that as voltage is stepped down current is stepped up. The ratio of the number of turns on each coil, called the turn's ratio, determines the ratio of the voltages. A step-down transformer has a large number of turns on its primary (input) coil which is connected to the high voltage mains supply, and a small number of turns on its secondary (output) coil to give a low output voltage. Turns ratio= $V_p/V_s=N_p/N_s$ and Power out=Power in.

Each of the blocks is described in more detail below,

- **Transformer** - steps down high voltage AC mains to low voltage AC.
- **Rectifier** - converts AC to DC, but the DC output is varying.
- **Smoothing** - smooths the DC from varying greatly to a small ripple.
- **Regulator** - eliminates ripple by setting DC output to a fixed voltage.

Power supplies made from these blocks are described below with a circuit diagram and a graph of their output.

- Transformer + Rectifier
- Transformer + Rectifier + Smoothing
- Transformer + Rectifier + Smoothing + Regulator

1. Transformer + Rectifier + Smoothing + Regulator

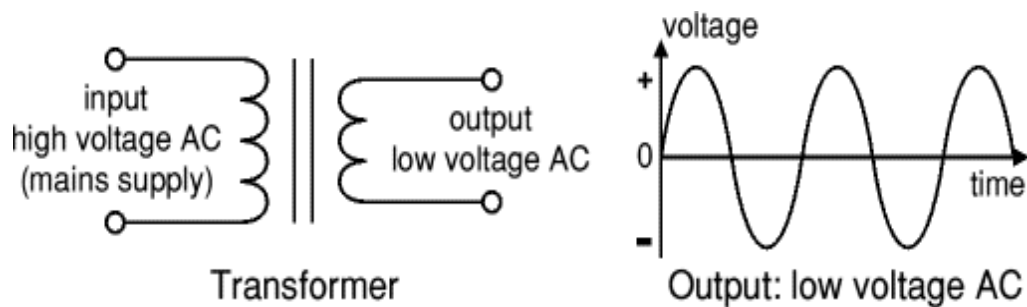


Figure 6.4 AC to DC Power Supply

The transformer converts high voltage AC from the mains supply into low voltage AC. This step-down process ensures safety and makes the voltage suitable for electronic circuits. However, the output is still alternating current (AC), which is not directly usable for most electronic devices. Therefore, further stages like rectifier, smoothing, and regulator are required to convert it into stable DC voltage.

RECTIFIER

There are several ways of connecting diodes to make a rectifier to convert AC to DC. The bridge rectifier is the most important and it produces full-wave varied DC. A full wave rectifier can also be made from just two diodes if a Centre-tap transformer is used, but this method is rarely used now that diodes are cheaper.

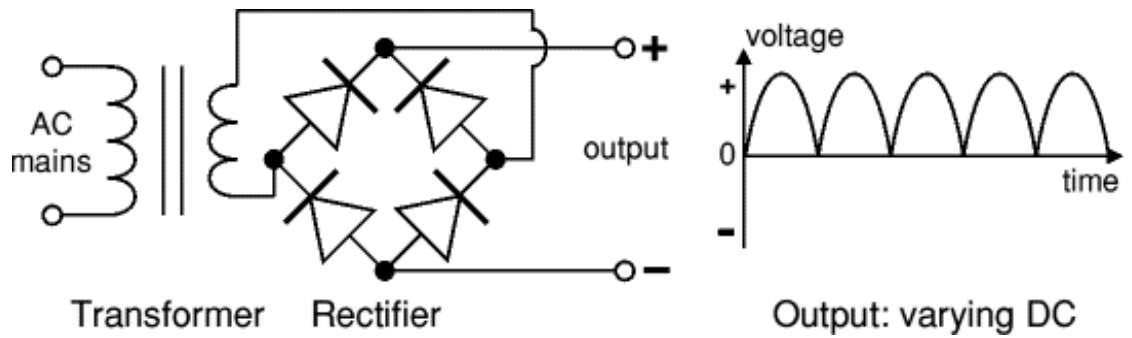


Figure 6.5 Full Wave Rectifier with Transformer

The AC mains supply is first given to a transformer, which adjusts the voltage level. The transformed AC signal is then fed into a bridge rectifier consisting of four diodes. During both the positive and negative half cycles of the AC input, the diodes conduct in such a way that the current flows in the same direction through the load. As a result, the output obtained is a pulsating DC voltage. The waveform on the right side represents this varying DC output, where all negative portions are converted into positive. This type of rectifier is widely used in power supply circuits because of its high efficiency and better output compared to half-wave rectifiers.

2. Transformer + Rectifier + Smoothing

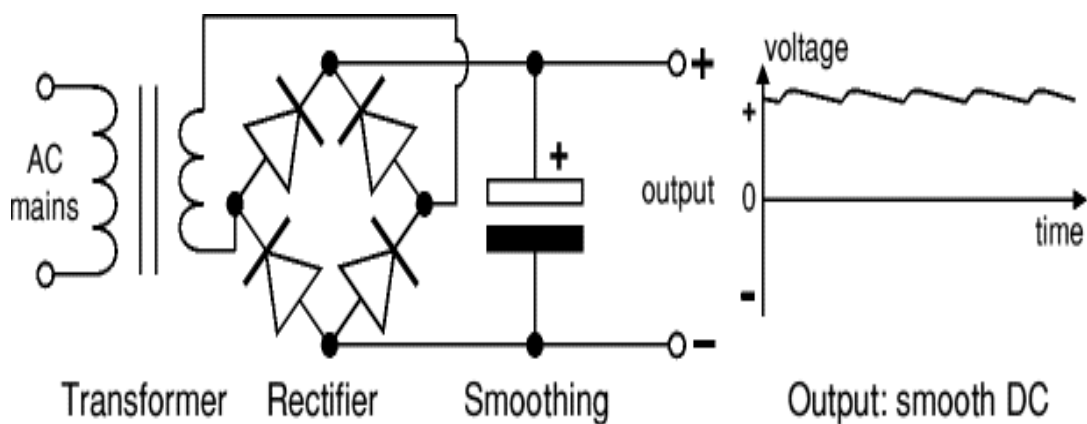


Figure 6.6 Full Wave Rectifier with Capacitor Filter

The AC mains supply is first applied to a transformer to adjust the voltage level. The transformer output is then given to a bridge rectifier made of four diodes, which converts the AC input into a pulsating DC signal. This capacitor charges during the peak voltage and discharges slowly when the voltage drops, thereby smoothing the output waveform. As a result, the final output is a nearly constant DC voltage with very small ripples.

3. Transformer + Rectifier + Smoothing + Regulator

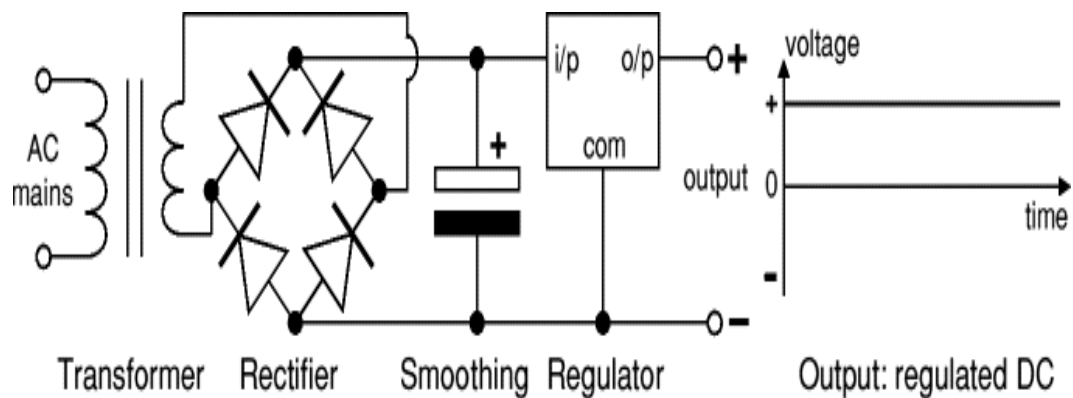


Figure 6.7 Regulated DC Power Supply using Transformer

The AC mains input is first applied to a transformer, which steps up or steps down the voltage to the required level. The transformer output is then fed into a bridge rectifier consisting of four diodes, which converts the AC voltage into pulsating DC. A smoothing capacitor is used next to reduce the ripple content and produce a smoother DC signal. Finally, a voltage regulator is connected to maintain a constant output voltage regardless of variations in input or load conditions. The resulting output is a stable and regulated DC voltage, as shown in the waveform. This type of power supply is widely used in electronic circuits for reliable and consistent performance.

Transformers waste very little power so the power out is (almost) equal to the power in. Note that as voltage is stepped down current is stepped up. The ratio of the number of turns on each coil, called the turns ratio, determines the ratio of the voltages. A step-down transformer has many turns on its primary (input) coil which is connected to the high voltage mains supply, and a small number of turns on its secondary (output) coil to give a low output voltage.

Formula Used,

Turns ratio= $V_p/V_s=N_p/N_s$ and Power Out($V_s \times I_s$)=Power In($V_p \times I_p$)

V_s = secondary(output)voltage

N_s = number of turns on secondary

coil I_s = secondary (output) current

V_p = primary(input)voltage

N_p =number of turns on primary

coil I_p = primary (input) current

Voltage Regulator

Voltage regulators and protection circuits are incorporated to prevent over-voltage, under-voltage, and short-circuit conditions, ensuring safe and reliable operation. Efficient power management also helps extend battery life, enabling prolonged use without frequent recharging. The stable power supply contributes to consistent sensor performance and uninterrupted wireless communication, making the device dependable for continuous intra-ocular pressure monitoring and remote healthcare applications.

They play a crucial role in ensuring the stable operation of sensitive components. For example, in computer power supplies, voltage regulators stabilize the DC voltages required by the processor, memory, and other internal components, ensuring reliable performance and preventing damage due to voltage fluctuations. In automobile alternators and central power station generator plants, regulators control and maintain the output voltage to ensure consistent power delivery.

Similarly, in electric power distribution systems, voltage regulators are installed at substations or along distribution lines to guarantee that all consumers receive a steady voltage regardless of changes in load demand. A typical voltage regulator circuit consists of four main parts.

The first is the reference voltage circuit, which provides a stable reference level for comparison. The second is the error amplifier, which compares the output voltage with the reference voltage and detects any deviation. The third component is the series pass transistor, which regulates the amount of current flowing to the load based on the error

signal. Finally, the feedback network continuously monitors the output voltage and feeds it back to the system, ensuring precise regulation and maintaining a constant output under varying conditions.

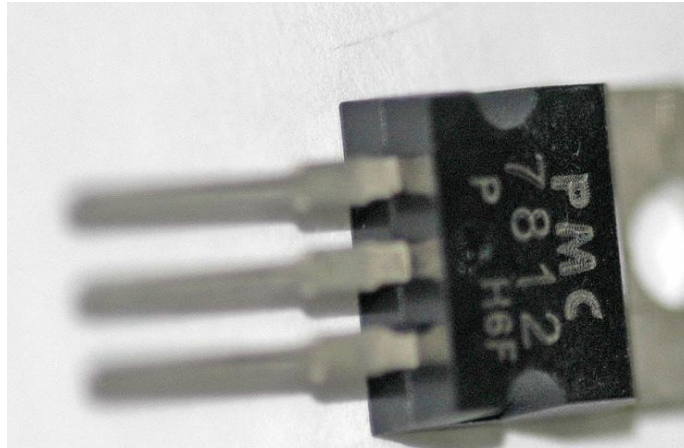


Figure 6.8 Voltage Regulator

6.5 ESP32 MICROCONTROLLER

The ESP32 microcontroller serves as the central processing and control unit of the proposed IoT-based smart tonometer system, ensuring efficient coordination of all hardware and software components. It is chosen for this application due to its powerful dual-core processor, low energy consumption, and built-in WIFI and Bluetooth capabilities, which make it highly suitable for portable and real-time healthcare solutions. The ESP32 plays a crucial role in acquiring data from the infrared or optical sensor that detects corneal deformation during the air-puff measurement process. Once the sensor captures the deformation signal, the ESP32 processes the raw data through embedded algorithms, including noise filtering and signal conditioning, to improve accuracy.

It then applies AI-based models to predict intra-ocular pressure and assess the risk of glaucoma. The microcontroller also controls peripheral components such as the OLED display, presenting real-time results to the user in a clear and user-friendly manner. Additionally, the ESP32 enables seamless IoT connectivity by transmitting the processed data to cloud platforms or mobile applications via Wi-Fi or Bluetooth. This allows remote monitoring by healthcare professionals, supports data logging, and facilitates early diagnosis.

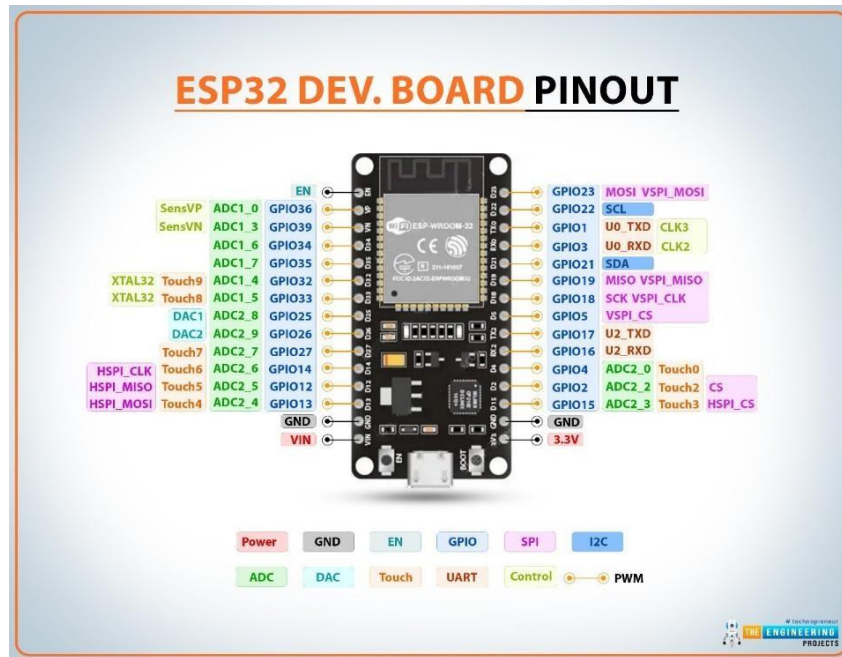


Figure 6.10 ESP32 Microcontroller

NodeMCU ESP8266

NodeMCU is an open source IoT stage. It incorporates firmware which keeps running on the ESP8266 Wi-Fi SoC from Expressive Systems, and equipment which depends on the ESP-12 module. The term NodeMCU typically refers to the firmware, whereas the board is termed Devkit. NodeMCU Devkit 1.0 consists of associate ESP-12E on a board that facilitates its use. It Additionally, it contains a transformer, a USB interface. The firmware utilizes the Lua scripting dialect. The NodeMCU (Node Micro Controller Unit) is an open-source software and hardware development environment that is built around a very inexpensive System-on-a-Chip (SoC) called the ESP8266.

NodeMCU is a widely used development platform for Internet of Things applications due to its low cost, compact size, and built-in Wi-Fi capability. It provides low-level hardware control through firmware, allowing efficient interaction with sensors and other electronic components. NodeMCU is based on the ESP8266 Wi-Fi System-on-Chip (SoC), particularly the ESP-12 module, which integrates processing and communication.

It uses the TCP/IP protocol stack to establish communication with the internet through a router. It supports IEEE 802.11 b/g/n standards, ensuring stable and efficient wireless data transmission. The module is powered by a Tensilica L106 32-bit RISC microprocessor, along with sufficient memory to handle real-time applications and data processing tasks. For interfacing, ESP8266 can relate to microcontrollers like Arduino using pins such as GND, TX, RX, and 3.3V. To program it using Arduino IDE, the ESP8266 board package must be installed through the Board Manager. NodeMCU provides 12 GPIO pins and one ADC pin. In this project, 7 GPIO pins (D0 to D6) are used for digital input and output operations, enabling efficient control and data acquisition.

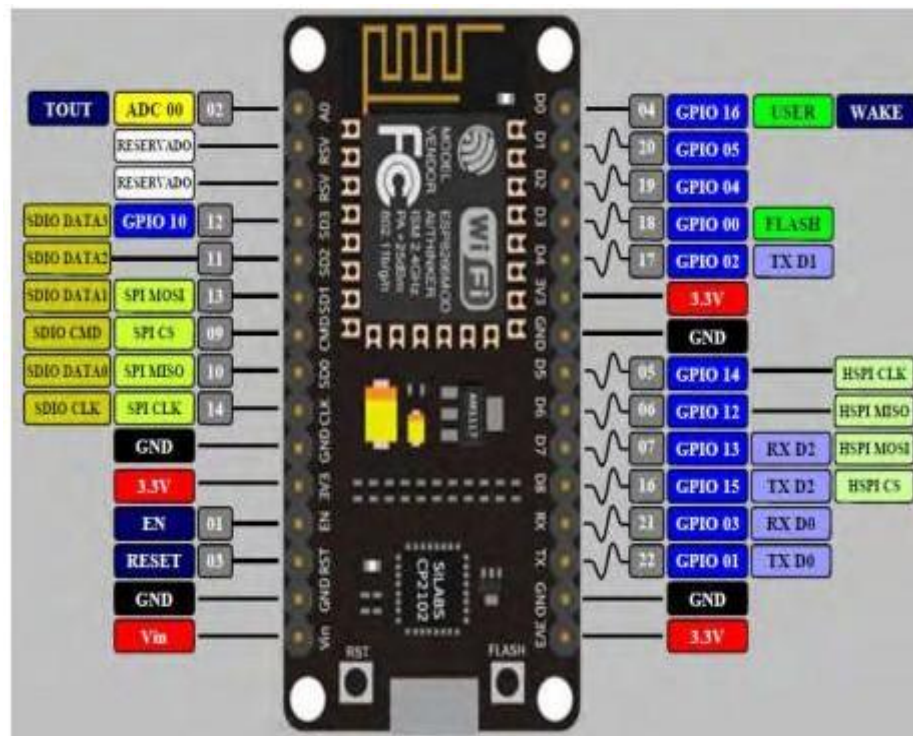


Figure 6.11 ESP8266 NodeMCU Chip

6.6 OLED DISPLAY MODULE

The OLED (Organic Light Emitting Diode) display module is used in the IoT Smart Tonometer to provide local visual feedback to the user. It displays important

information such as measured intra-ocular pressure values, system status, connectivity indication, and alert messages. The OLED display is chosen because of its high contrast, low power consumption, and compact size, making it suitable for portable healthcare devices. In this system, the OLED display is interfaced with the ESP32 microcontroller using standard communication protocols such as I²C or SPI. The microcontroller updates the display in real time after each measurement cycle, allowing users to instantly view their eye pressure readings without relying solely on a mobile application. This feature is especially beneficial for elderly users and for quick on-device verification.



Figure 6.12 OLED Display

Unlike traditional LCDs, OLED displays do not require a separate backlight, as each pixel emits its own light. this significantly reduces power consumption, making the system more energy-efficient and suitable for battery-operated portable medical devices like the smart tonometer. additionally, OLED displays offer high contrast ratios and deep blacks, which enhance visibility even in low-light or bright ambient conditions.

The OLED display module plays a vital role in enhancing user experience, ensuring proper device operation, and improving the overall reliability of the smart tonometer system.

6.7 WI-FI COMMUNICATION INTERFACE

The Wi-Fi communication interface plays a vital role in enabling the IoT Smart Tonometer to transmit intra-ocular pressure data wirelessly to a cloud platform for real-time monitoring and detailed analysis. In this system, Wi-Fi connectivity is efficiently achieved using the built-in Wi-Fi module of the ESP32 microcontroller. This integration removes the need for any additional external communication hardware, thereby reducing system complexity, cost, and power consumption while improving overall reliability. After collecting data from the spectral imaging sensor, AI-based processing techniques are applied to accurately predict intra-ocular pressure values. Furthermore, the Wi-Fi module is capable of sending instant alert notifications whenever abnormal intra-ocular pressure levels are detected, helping in early diagnosis of conditions like glaucoma. This wireless communication feature significantly enhances the portability and usability of the device, making it highly suitable for home-based healthcare and tele-ophthalmology applications.

6.8 OVERALL HARDWARE INTEGRATION

The overall hardware integration of the IoT Smart Tonometer is carefully designed to ensure seamless interaction among all system components while maintaining compact size, portability, and high reliability. This integration plays a crucial role in achieving accurate, consistent, and non-invasive intra-ocular pressure measurement, which is essential for early detection and continuous monitoring of eye-related diseases such as glaucoma. A well-coordinated hardware architecture ensures that each component functions efficiently without interference, thereby enhancing the overall system performance. This step significantly improves the overall accuracy and reliability of the system. Once the signals are properly conditioned, they are fed into the ESP32 microcontroller, which serves as the central processing unit of the system. The ESP32 performs multiple critical tasks, including sensor interfacing, timing control, and real-time data acquisition.

CHAPTER 7

SYSTEM DESIGN

7.1 CIRCUIT DIAGRAM

This circuit diagram represents the hardware implementation of the IoT Smart Tonometer system. The ESP32 microcontroller acts as the central processing unit, interfacing with the spectral sensing module and managing data acquisition. An OLED display is connected via I2C communication to provide real-time intra-ocular pressure readings and system status. The sensor module captures optical data related to corneal deformation, which is processed by the ESP32. A regulated power supply, supported by a rechargeable battery, ensures stable operation. The circuit enables efficient sensing, processing, and wireless data transmission, making it suitable for portable and continuous glaucoma monitoring applications.

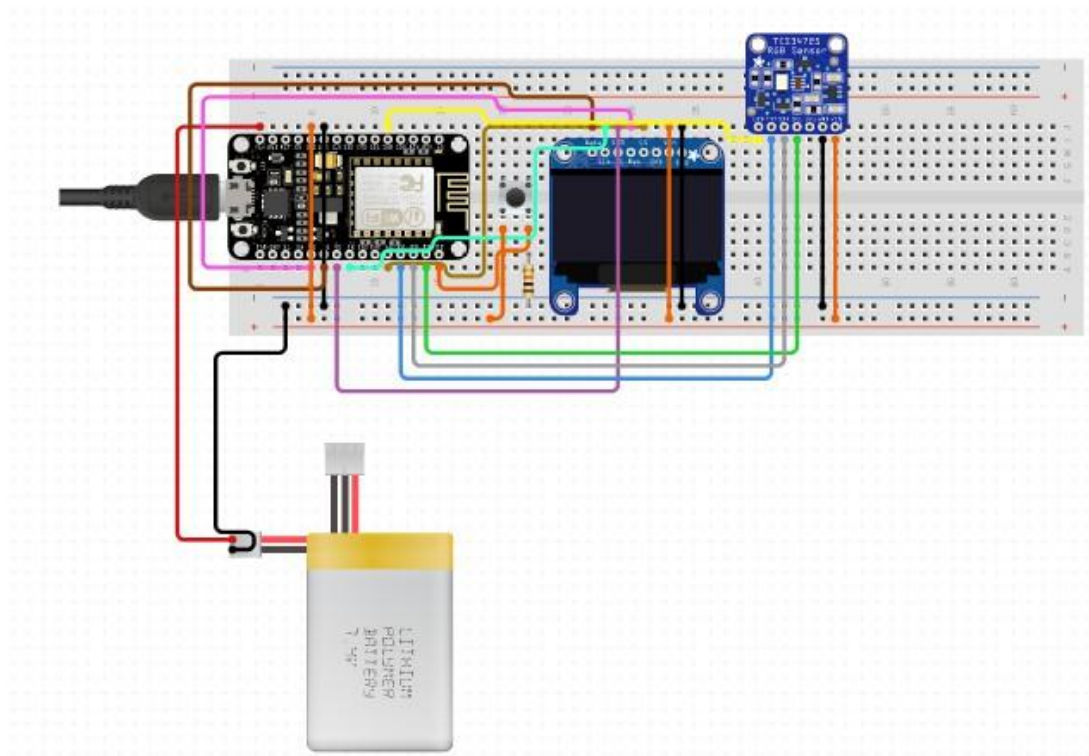


Figure 7.1 Prototype Circuit Design for AI-Based IoT Smart Tonometer

7.2 WORKING

The IoT Smart Tonometer operates on the principle of measuring corneal deformation in response to a controlled, non-invasive stimulus and estimating intra-ocular pressure using advanced AI-based analysis. This system combines spectral imaging technology, embedded processing, and wireless communication to deliver accurate, reliable, and continuous IOP monitoring beyond traditional clinical settings. As a result, it is highly suitable for home-based healthcare, tele-ophthalmology applications, and early detection of glaucoma, significantly reducing the need for frequent hospital visits and enabling proactive eye care management. The measurement process begins when the device is powered on and the user positions their eye correctly in front of the sensing unit. Proper alignment is critical for accurate readings, and the system may include visual indicators, guiding lights, or mechanical supports to help the user maintain the correct position. Once aligned, a gentle air-puff or rebound mechanism applies a short, safe, and non-contact force to the cornea. This force induces a slight deformation in the corneal surface. The degree of this deformation is inversely related to the intra-ocular pressure inside the eye-eyes with higher IOP exhibit less deformation due to increased stiffness, while those with lower IOP deform more easily.

During this deformation event, a spectral imaging sensor captures detailed optical responses of the corneal surface across multiple wavelengths. Unlike conventional tonometer's that relay on single-point measurements, spectral imaging provides a richer and more comprehensive dataset. It analyses variations in reflected light intensity and wavelength distribution, allowing the system to detect subtle deformation patterns with high precision. This multi-wavelength approach enhances sensitivity and ensures reliable performance even under challenging conditions such as slight eye movements, blinking, or changes in ambient lighting.

The raw data collected from the spectral sensor is then passed through a signal conditioning and noise filtering stage. This module plays a crucial role in improving data quality by removing unwanted disturbances such as electronic noise, motion artifacts, and environmental interference. The cleaned analog signals are then converted into digital form for further processing.

These digital signals are transmitted to the ESP32 microcontroller, which acts as the central control and processing unit of the system. The ESP32 efficiently manages sensor interfacing, timing control, and data acquisition while maintaining low power consumption. Once the data is processed, it is fed into an AI-based intra-ocular pressure prediction model. This model is trained using datasets that establish the complex and nonlinear relationship between corneal deformation characteristics and actual IOP values.

By extracting meaningful features from the spectral data, the AI algorithm accurately predicts the intra-ocular pressure. This intelligent processing approach significantly improves measurement accuracy and consistency compared to traditional tonometry techniques, which may be affected by user handling or environmental conditions. After the IOP value is estimated, the result is immediately displayed on an OLED display module. The display provides clear, high-contrast, and real-time feedback to the user, including IOP readings, system status, and possible alerts. This ensures that users can quickly understand their eye health condition without requiring professional assistance.

At the same time, the ESP32 utilizes its integrated Wi-Fi capability to transmit the data securely to a cloud platform. The cloud system stores historical measurement data, enabling long-term tracking and trend analysis of intra-ocular pressure. Healthcare professionals can remotely access this data through web or mobile applications, facilitating continuous patient monitoring and timely medical intervention. If abnormal IOP levels or sudden fluctuations are detected, the system can automatically generate alerts and notifications, prompting immediate attention. Overall, the IoT Smart Tonometer represents a portable, efficient, and intelligent solution for continuous eye pressure monitoring. By integrating advanced sensing technologies, AI-driven analysis, and IoT connectivity, the system enhances accessibility to eye care services and supports early diagnosis and prevention of serious conditions like glaucoma.

7.3 CNN Algorithm

In the proposed IoT-based Smart Tonometer system, the Convolutional Neural Network plays a crucial role in accurately predicting intraocular pressure by analyzing corneal deformation data obtained from the sensing unit.

CNN is specifically chosen for its ability to automatically extract meaningful features from complex input data, making it highly suitable for medical signal and image-based analysis. In this project, the input to the CNN model consists of processed signals or spectral imaging data that represent the corneal response to the applied air-puff or rebound stimulus. Initially, the raw data captured by the sensor undergoes preprocessing steps such as noise filtering, normalization, and signal enhancement. This ensures that the input to the CNN is clean, consistent, and suitable for analysis.

The pre-processed data is then structured into a format compatible with the CNN, such as a one-dimensional signal array or two-dimensional feature map, depending on the type of sensor output. CNN begins its operation with convolutional layers, where multiple filters are applied to the input data to extract important features such as peaks, edges, and deformation patterns. These filters help in identifying subtle variations in corneal behaviour that are directly related to intraocular pressure.

Following the convolutional layers, pooling layers are used to reduce the dimensionality of the feature maps while preserving the most significant information. This not only reduces computational complexity but also improves the model's ability to generalize. The extracted features are then passed through fully connected layers, where the CNN learns the nonlinear relationship between corneal deformation characteristics and actual IOP values. During the training phase, the model is trained using labelled datasets containing known IOP values, allowing it to adjust its internal parameters and improve prediction accuracy.

Once trained, the CNN model is deployed within the system, where it processes real-time input data from the sensor. The model quickly analyses the features and outputs an accurate IOP prediction. This prediction is then displayed to the user and transmitted to the cloud for further monitoring. The use of CNN enhances the system's robustness against noise, eye movement, and environmental variations. Overall, the CNN algorithm significantly improves the intelligence, accuracy, and reliability of the smart tonometer system, making it highly effective for early glaucoma detection and continuous eye health monitoring.

CHAPTER 8

CONCLUSION & FUTURE ENHANCEMENT

8.1 CONCLUSION

The project presents the design and implementation of an IoT-based Smart Tonometer for AI-based intra-ocular pressure prediction and early glaucoma detection. The main goal of this system is to provide a portable, affordable, and user-friendly solution for monitoring eye pressure outside traditional clinical environments. Conventional tonometers are often expensive, bulky, and limited to hospital use, making continuous monitoring difficult for many patients. This proposed system addresses these challenges by integrating advanced sensing, embedded processing, artificial intelligence, and IoT connectivity into a compact device. The system uses a spectral imaging sensor to capture subtle corneal deformation characteristics caused by controlled stimulation of the eye surface. These deformation patterns are closely related to intra-ocular pressure levels. The sensor data is processed by an ESP32 microcontroller, which performs signal conditioning, noise filtering, and feature extraction to ensure reliable measurements. An embedded AI prediction model then analyses the processed data to estimate the IOP value with improved accuracy.

One of the key advantages of the system is the integration of AI-based compensation techniques that reduce measurement errors caused by blinking, minor eye movements, and sensor drift. This improves the reliability of the readings compared to traditional single-measurement tonometers. The measured results are displayed in real time through an OLED display, allowing users to quickly check their eye pressure status. In addition, the IoT capability enables cloud connectivity, allowing patient data to be stored and accessed remotely by doctors or healthcare providers. This supports continuous monitoring, remote diagnosis, and long-term IOP trends. Overall, the proposed IoT Smart Tonometer offers a cost-effective, portable, and scalable solution that can significantly support early glaucoma detection and improve eye health monitoring, especially for elderly individuals and patients living in remote or underserved areas.

8.2 FUTURE ENHANCEMENT

The proposed IoT Smart Tonometer system can be further enhanced through several advancements in both hardware and software to improve performance, usability, and clinical applicability. One major improvement is the integration of advanced deep learning models, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to enhance prediction accuracy. These models can be trained on diverse datasets that include variations in age, corneal thickness, and eye conditions, enabling the system to deliver more personalized and reliable intra-ocular pressure estimations across different patient profiles. Another important enhancement is the miniaturization of hardware components. By reducing the size of sensors, processing units, and optical assemblies, the device can be transformed into a more compact and lightweight form, potentially evolving into a wearable solution. This would significantly improve user comfort and enable continuous or frequent monitoring without inconvenience. The addition of a dedicated mobile application with multilingual support can greatly enhance accessibility and user engagement. Such an application can provide real-time data visualization, alerts, historical trends, and easy communication with healthcare providers. Multilingual interfaces ensure that users from different regions and backgrounds can effectively interact with the system. To further improve usability, real-time video guidance can be incorporated into the system. This feature would assist users in correctly positioning their eyes during measurement, reducing errors caused by improper alignment and improving overall accuracy. Power management can also be optimized by incorporating battery health monitoring and ultra-low power operating modes. These features will extend device lifespan, ensure safe operation, and make the system more reliable for long-term use, especially in remote or resource-limited settings. This ensures that healthcare professionals have immediate access to accurate and up-to-date patient information. Finally, the system can be expanded to detect additional ocular conditions by analysing extended spectral data patterns.

CHAPTER 9

APPENDIX

9.1 SOURCE CODE

```
#include

"Arduino.h"

#include "Wire.h"

#include "SPI.h"

#include "Adafruit_SSD1306.h"

#include "Adafruit_GFX.h"

#include "Button.h"

// Pin Definitions

#define OLED128X64_PIN_RST      15

#define OLED128X64_PIN_DC      12

#define OLED128X64_PIN_CS      2

#define PUSHBUTTON_PIN_2       16

#define TCS34725_PIN_LED        9

#define TCS34725_PIN_INT 0

// object initialization

#define SSD1306_LCDHEIGHT 64

Adafruit_SSD1306      oLed128x64(OLED128X64_PIN_DC,
                                   OLED128X64_PIN_RST, OLED128X64_PIN_CS);

Button pushButton(PUSHBUTTON_PIN_2);

// define vars for testing menu
```

```

const int timeout = 10000;          //define timeout
of 10 sec

/ Setup the essentials for your circuit to work. It runs first every time your circuit is
powered with electricity.

void setup()

    // Setup Serial which is useful for debugging

    // Use the Serial Monitor to view printed messages

    Serial.begin(9600);

    while (!Serial) ; // wait for serial port to connect. Needed for native USB

    Serial.println("start");

    oLed128x64.begin(SSD1306_SWITCHCAPVCC); // by default, we'll
rate the high voltage from the 3.3v line internally! (neat!)

    oLed128x64.clearDisplay(); // Clear the buffer.

    oLed128x64.display();

    pushButton.init();

    menuOption =

    menu();

}

void loop()

{

    if(menuOption == '1') {

        // Monochrome 1.3 inch 128x64 OLED graphic display - Test Code

```

gene

```

oLed128x64.setTextSize(1);

oLed128x64.setTextColor(WHITE)

; oLed128x64.setCursor(0, 10);

oLed128x64.clearDisplay();

oLed128x64.println("Circuito.io Rocks!"); oLed128x64.display();

delay(1);

oLed128x64.startscrollright(0x00,

0x0F); delay(2000);

oLed128x64.stopscroll();

delay(1000);

oLed128x64.startscrollleft(0x00,

0x0F); delay(2000);

oLed128x64.stopscroll();

}

else if(menuOption == '2') {

bool pushButtonVal = pushButton.read();

Serial.print(F("pushButtonVal: ")); Serial.println(pushButtonVal);

}

else if(menuOption == '3')

{

}

char menu()

{

```

```

Serial.println(F("\nWhich component would you like to test?"));

Serial.println(F("(1) Monochrome 1.3 inch 128x64 OLED graphic
display")); Serial.println(F("(2) Mini Pushbutton Switch"));

// Read data from serial monitor if received while
(Serial.available())

    if(c == '1')

        Serial.println(F("Now Testing Monochrome 1.3 inch 128x64 OLED
graphic display"));

        else if(c == '2')

            Serial.println(F("Now Testing Mini Pushbutton
Switch")); else if(c == '3')

else

    {

        Serial.println(F("illegal
input!")); return 0;

    }

    time0 = millis();

    return c;

}

}

```

9.2 OUTPUT MODEL

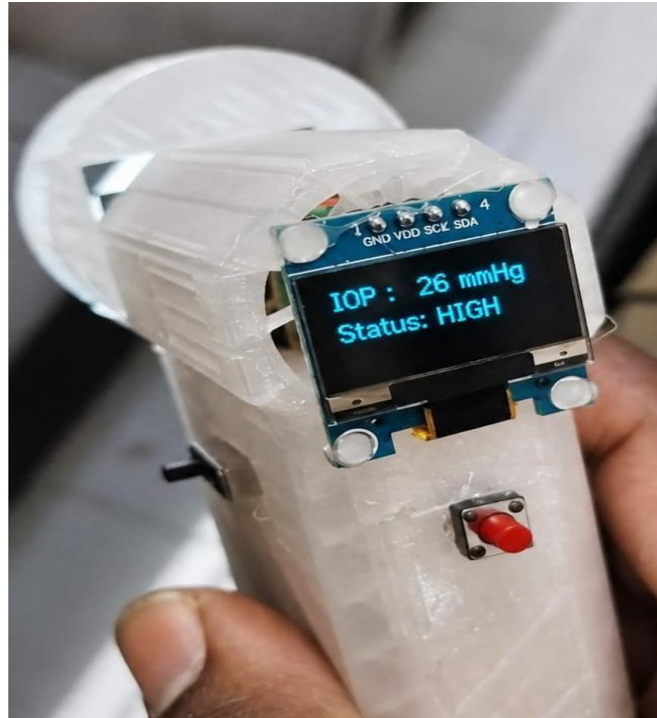


Figure 9.1 IOP Value Display on OLED



Figure 9.2 Mobile Alert Notification

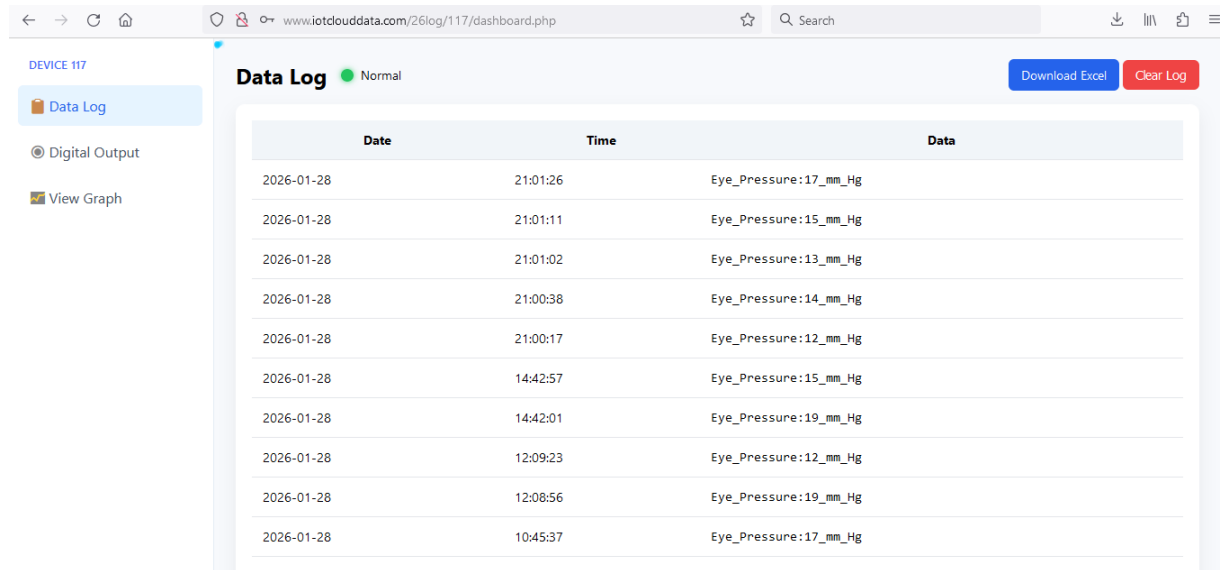


Figure 9.3 Web Page

Live Data Graph

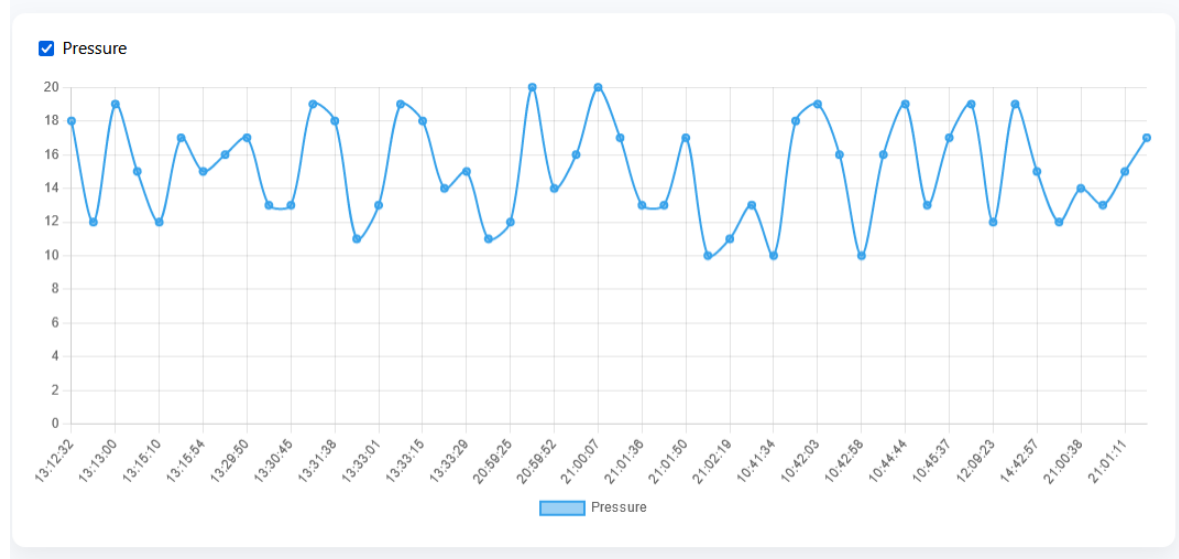


Figure9.4 Live Graph

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